

Competitive Dynamics in Organizations

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Abstract

Political economy has studied how institutions divide an already-produced surplus, but has discussed far less about how that surplus is politically determined in the first place. We develop a political theory of production: economic performance depends on how organizations secure political control for compliance within hierarchies. When legal enforcement is strong, authority is predominantly contractual and merit-based competition is strengthened. When legal enforcement is weak, authority relies more heavily on individualized loyalty, creating a trade-off between preserving favoritism and sustaining high-powered incentives. Because a large loyal following lets a superior sustain intense competition even without legal backing, weak law does not mechanically depress productivity, which explains why higher performance can be achieved in nondemocratic countries and productivity divergence within and across regime types. The same mechanism generates political inequality, as effort gaps between loyal and disloyal subordinates emerge from authority-preserving competition design. Extending the model to two periods, we show that superiors have a standing incentive to expand their authority capital, and that this accumulation eventually makes weakening legal enforcement privately optimal even when strong enforcement remains socially efficient, a structural account of institutional backsliding.

Keywords: authority capital; legal enforcement; incentive design; productivity divergence; economic inequality

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Introduction

Political economy offers many theories explaining why economic performance varies within and across regime types. For instance, work on electoral competition and constitutional design explains variation in the composition of public spending and the extent of rent extraction (Persson et al., 2000); work on clientelism explains why patronage crowds out efficient public provision even under robust electoral competition (Robinson and Verdier, 2013); and selectorate theory links the public–private mix of goods provision to the size of a leader’s winning coalition (Bueno de Mesquita et al., 2003).

These theories have provided valuable insights into how political institutions shape economic outcomes. Yet they largely approach economic outcomes through the politics of redistribution: how an already-produced surplus is divided among public goods, patronage, targeted transfers, or rents. They tell us much about how political institutions “allocate the cake”, but little about how “the cake is politically produced” before the redistribution can be realized. To understand how political mechanisms shape productivity within and across regime types, we need not only a politics of distribution, but also a political theory of production—one that treats economic performance as an *ex ante* incentive problem rooted in the political conditions of production inside organizations. How, then, do political institutions shape the incentive to produce—and why does this vary not only within regime types, but across them?

Production requires not only economic incentives but also political control capable of sustaining compliance and stability (Polanyi, 1944). Yet political control does not sustain itself. Across organizations, from states and bureaucracies to firms, a common way for superiors to secure it is to place loyal subordinates in key positions. In a hierarchical structure where subordinates far outnumber key positions, promotion typically creates internal competition based on relative performance. Favoritism toward loyalists, however, may condition this competition, alter subordinates’ effort, and ultimately shape organizational output. This paper formalizes this tension in a model of competitive dynamics across institutional contexts.

The model is set in a principal-agent framework that extends Berry’s (1993) contest framework to incorporate political control. An organization consists of three tiers: a superior at the top (e.g., a political leader, a bureaucratic chief, or a firm’s CEO), a single position or reward at the middle, and multiple subordinates at the bottom. Subordinates are identical in their work ability but heterogeneous in loyalty, defined as their propensity to comply with the superior’s directives once promoted. Subordinates exert costly effort to compete for the middle position, and the superior sets the intensity of this competition by weighing loyalty.

The extent to which this control depends on loyalty varies with the institutional environment in which the organization operates. Where the rule of law and formal

institutions are stronger, organizations rely less on informal institutions such as personal loyalty to secure compliance, because legal enforcement and enforceable contracts can guarantee it directly; where legal enforcement is weaker, dependence on loyalty is greater. A second, independent factor is the superior's authority capital—the number of subordinates already loyal to her. Together, these two factors determine how intensely the superior runs the competition for the position, and the level of effort this competition elicits from subordinates.

Organizational productivity remains high through two distinct mechanisms, when either legal enforcement or the superior's authority capital is strong. When legal enforcement is strong, the superior relies less on personal loyalty to secure compliance and can therefore sustain promotion based primarily on relative performance. More meritocratic competition strengthens subordinates' incentives to exert productive effort, generating higher aggregate output. When legal enforcement is weak but authority capital is high, intense competition can be sustained for a different reason. Because loyalists constitute a large share of the subordinate pool, a meritocratic winner is already likely to be loyal. The superior can therefore maintain intense performance-based competition without substantially jeopardizing political control. High authority capital thus allows productive competition and political control to coexist even where formal institutions are weak.

Organizational productivity declines only when legal enforcement and the superior's authority capital are both weak. Under these conditions, the superior cannot rely on either formal institutions or an established network of loyal subordinates to secure compliance. She therefore relaxes performance-based competition and shifts promotion away from merit and toward loyalty. This shift weakens subordinates' incentives to exert productive effort, ultimately reducing aggregate organizational output.

This mechanism is consistent with broad historical and contemporary patterns: strong rule-of-law institutions are one foundation for high-powered incentives, but not the only one. In several historical cases, including the Ottoman Empire's centralized fiscal-administrative system (Karaman and Pamuk, 2010; Pamuk, 2020), Meiji Japan's state-led modernization (Johnson, 1982), Prussia's administrative and market-oriented reforms ((Dincecco, 2011; Andersen, 2018), and imperial China's bureaucratic coordination (Chen, 2024; Wang, 2022), large-scale economic mobilization emerged before modern legal institutions were fully developed, supported instead by centralized political control and bureaucratic capacity. The same logic holds in modern settings: China's reform-era promotion system tied local officials' careers to economic performance while maintaining loyalty as the basis of authority. Factional ties to central leaders have often outweighed growth performance in shaping advancement, and this reliance on loyalty grows stronger at higher levels of the hierarchy (Shih et al., 2012). At the firm level, family CEO succession (Pérez-González, 2006; Bennedsen et al., 2006)

and cross-country differences in management practices (Bloom and Van Reenen, 2007, 2010) show that allocation rules based on family ties strongly shape productivity.

The heterogeneous incentives that subordinates face, shaped by favoritism and by authority capital, are a key factor not only in generating aggregate productivity differences across institutional contexts but also in shaping broader economic inequality in our model. When competition weakens, equilibrium effort decreases for all subordinates, but loyal subordinates choose higher effort than disloyal subordinates, because loyal subordinates have a weakly higher probability of being promoted to key positions or receiving valuable organizational resources. This generates heterogeneity in effort and reward across loyal and disloyal subordinates. Such politically induced differences in incentives constitute a central mechanism through which political control generates economic inequality. This mechanism offers a theoretical account of why, within China's political hierarchy, city leaders with strong ties to their provincial patron have been found to outperform their unconnected counterparts (Jiang, 2018).

In the model extension, we investigate how the superior's payoff varies with authority capital and legal strength when both are allowed to change. We show that the superior's payoff is always weakly increasing in authority capital, reflecting a structural incentive for hierarchical organizations to accumulate authority capital. The relationship between legal strength and the superior's payoff, by contrast, is not monotonic: when authority capital is limited, strengthening legal enforcement beyond a certain point initially lowers the superior's payoff, and only raises it once legal strength crosses a threshold we call a tipping point. Below it, the superior still gains from defending the law; above it, she gains from letting it erode. As authority capital grows, this tipping point falls, so that a superior with a sufficiently large loyal following prefers to erode legal enforcement almost regardless of how strong that enforcement already is.

This paper contributes to three fields in political science. First, the theory provides a microfoundation for why economic performance diverges both within and across regime types. Our model identifies two more fundamental sources of variation—the strength of legal enforcement and the superior's authority capital—that jointly determine the intensity of performance-based competition an organization can sustain without jeopardizing political control. This framework reconciles conflicting evidence on the economic performance of democratic and authoritarian regimes. We argue that regime type alone offers an incomplete account of productivity differences; what matters is how political control structures incentives inside organizations.

Second, this paper proposes an alternative possibility about the relationship between economic development and inequality. The classic Kuznets hypothesis conceptualizes development and inequality as systematically linked, with inequality first rising and then falling as development proceeds (Kuznets, 1955, 1966). Acemoglu and Robinson (2002) politicize this relationship, showing that different political trajec-

ries can generate different development–inequality paths. Unlike their account, our model shows that there need not be a stable relationship at all: high productivity can coexist with either low or high inequality, and low productivity can do the same.

Third, this paper pushes forward Helmke and Levitsky’s 2012 classic typology of how informal institutions coexist with formal ones. They sort informal institutions into a set of fixed types, whereas this paper shows that the “type” itself is not fixed: the same informal institution (authority capital, i.e. the loyalty network) can, through its own accumulation, shift from complementing formal legal enforcement to competing with it. As authority capital grows, the range of legal environments in which weakening the law is privately optimal weakly expands, and for sufficiently large authority capital, the superior retains almost no incentive to defend the law at all.

The paper proceeds as follows. Section 2 discusses related literature. Section 3 presents the baseline model, in which legal strength and authority capital jointly determine the superior’s optimal competition rule and the resulting patterns of productivity and political inequality. Section 4 extends the model to endogenize institutional change. Section 5 draws out the model’s implications for productivity, inequality, and institutional stability. Section 6 concludes.

Related Literature

Our paper builds on the organizational-economics literature on contests, tournaments, and promotion incentives. Classic tournament theory studies how relative performance evaluation and prize structures shape agents’ incentives to exert effort (Tullock, 2001; Lazear and Rosen, 1981; Rosen, 1986), while subsequent work develops contest success functions for environments with multiple prizes (Berry, 1993; Skaperdas, 1996). Related work in personnel economics emphasizes promotion and career concerns as mechanisms for both eliciting effort and selecting high-performing employees within organizations (Gibbons and Murphy, 1992; Prendergast, 1999; Baker et al., 1994). This literature treats the competitive structure itself as fixed by productive considerations, leaving its political determinants outside the analysis. We instead endogenize the competitive structure as an organizational choice constrained by the superior’s need to maintain political control, and examine how the resulting competition shapes productive effort and organizational output.

A closely related literature studies the trade-off between loyalty and competence in political appointments. Egorov and Sonin (2011) show that rulers may sacrifice competence for loyalty because more competent advisers are better able to identify profitable opportunities for defection. Zakharov (2016) derives a similar trade-off from subordinates’ outside options: more competent officials have better prospects under alterna-

tive rulers and therefore weaker incentives to invest in the incumbent's survival. More recently, Dessein and Garicano (2025) embed the loyalty–competence trade-off in political tournaments, showing how competition can increase leaders' demand for loyal team members who are less likely to exit when political prospects deteriorate. Despite their different mechanisms, these accounts share a common focus: loyalty affects organizational performance primarily through the selection and retention of personnel with different levels of competence or talent.

Our framework shifts attention from the selection effect of loyalty to its incentive effect. We abstract from the loyalty–competence trade-off by allowing subordinates to be equally capable and instead ask how political considerations shape the competition through which they are selected. In particular, competition is not simply an exogenous environment in which leaders choose between loyal and talented agents; its intensity is itself an endogenous organizational choice shaped by the superior's need for political control. Although promotion determines who ultimately occupies the position, our central mechanism operates before selection occurs: favoritism toward loyalists changes the expected return to productive effort, thereby altering how intensely all subordinates compete and, ultimately, how much the organization produces. Loyalty can therefore impact organizational productivity even when it entails no sacrifice in the competence of the person selected.

Finally, our paper speaks to theories of institutional change. A large literature explains institutional transformation through shifts in the distribution of political power (Acemoglu et al., 2005) and through critical junctures that unsettle existing institutional arrangements (Collier et al., 1991; Capoccia and Kelemen, 2007). Other work emphasizes that institutional change can arise endogenously, without any external disruption. Greif and Laitin (2004), for example, show how institutions can generate processes that gradually alter the parameters sustaining their own equilibrium, while theories of gradual institutional change emphasize how ambiguities in institutional rules and their implementation create opportunities for internally generated transformation (Mahoney et al., 2010).

Our framework identifies a distinct mechanism of endogenous institutional change: the accumulation of authority capital, an informal institution in the sense of Helmke and Levitsky (2012). Rather than occupying a fixed position in their typology, authority capital shifts from complementing formal legal enforcement to competing with it as the stock of loyal subordinates grows. Existing institutions shape how superiors secure political control, but the exercise of that control can itself accumulate this stock. As it grows, it changes the superior's dependence on, and hence private valuation of, formal legal institutions. This relationship is therefore recursive rather than one-directional: institutions shape political control, and the authority capital accumulated under them can in turn reshape the incentive to preserve those same institutions.

The Model

Superiors who preside over hierarchies rarely command compliance by fiat. A party leader cannot force local cadres to implement policy faithfully; a bureau chief cannot compel careerist subordinates to execute directives against their own interest; a firm's CEO cannot simply order loyalty from division heads who control day-to-day operations. Compliance has to be secured, and it can be secured in essentially two ways. Where courts, audits, and enforceable employment contracts are credible, a superior can rely on formal rules: promotion tracks performance, and legal or bureaucratic enforcement backstops the arrangement even against subordinates who would rather shirk or defect once in office. Where such enforcement is weak, the superior has no choice but to substitute an informal guarantee for a formal one: she places people she already trusts—because of kinship, faction, or personal loyalty—into the positions from which compliance is ultimately extracted. This substitution is what recurs across the historical and contemporary cases discussed above, from the Ottoman and Prussian bureaucracies to promotion politics inside the Chinese party-state.

The difficulty is that loyalty and performance-based competition pull against one another. Favoring loyalists secures the superior's political control, but it blunts the very competition that elicits effort from everyone else. This section builds a model to trace exactly how a superior resolves that tension, and to show that her resolution depends on two political conditions: the strength of the legal environment surrounding the organization, and the size of the loyal following she has already built. Three ingredients do the work. First, subordinates differ only in loyalty, not in ability, and the superior can act on loyalty only within whichever pool of top performers she has already committed to considering—favoritism operates *through*, not instead of, competition. Second, the weight the superior places on loyalty is disciplined by the surrounding rule of law. Third, that weight interacts with how large a loyal following, or *authority capital*, the superior already commands. We formalize each in turn, characterize the resulting equilibrium, and then ask how the superior optimally designs the competition itself.

Set Up

Organizational Structure and Agent Types. Consider a hierarchical organization consisting of a superior, S , and $n \geq 2$ subordinates. The superior might be a political leader allocating provincial appointments, a bureaucratic chief controlling promotion within an agency, or a firm's CEO choosing who reports directly to her; in each case she presides over more capable people than she has positions to reward. The subordinates compete for a single indivisible prize, $V > 0$ —a promotion, a delegation of authority, or another valuable position—and the superior's central design choice is the

rule that governs who is eligible to receive it.

Subordinates are identical in productive ability and in the cost of exerting effort, but they differ along a dimension that matters for political control rather than for output: loyalty, defined as a subordinate's personal willingness to comply with the superior's directives once given delegated authority. We refer to subordinates of the two resulting types as loyal (l) and disloyal (d). Of the n subordinates, $m \in \{0, 1, \dots, n\}$ are loyal and the remaining $n - m$ are disloyal; we call m the superior's *authority capital*, since it indexes the loyal following she can already draw on and, other things equal, a larger m leaves her political position more secure. Both individual types and m are common knowledge; m is treated as exogenous until Section 4, where the superior can invest in building it.

The political stakes of the loyalty distinction are direct. A loyal subordinate who is promoted complies voluntarily with the superior's directives, which we normalize to a payoff of $+1$ for the superior. A disloyal subordinate offers less reliable compliance, imposing additional monitoring and enforcement costs that we normalize to -1 .

Legal Strength. How much this distinction matters to the superior is not fixed: it depends on the strength of the legal environment in which the organization is embedded. Where the rule of law is strong, impersonal and enforceable contracts can secure compliance directly, so the superior has comparatively little need to place loyalists in key positions. Where legal enforcement is weak, contractual guarantees lose their force, and personal loyalty becomes the superior's main available substitute for institutional guarantees of compliance. Formally, let $\alpha \geq 0$ denote the weight the superior places on loyalty, decreasing in the strength of the rule of law within which the organization is situated. Like m , α is treated as exogenously given in the baseline model.

The Eligibility Rule. Before subordinates act, the superior sets the rule governing who can receive the prize: an eligibility threshold

$$k \in \mathcal{K} \equiv \{1, \dots, n\}.$$

After effort is exerted, only the k highest-ranked subordinates are eligible to receive the prize. We refer to the inverse of k as *competition intensity*, since performance-based competition is more intense the smaller k is.

At one extreme, $k = 1$ is a strict tournament: only the highest-ranked subordinate is eligible, leaving the superior no discretion over who receives the prize. At the other extreme, $k = n$ makes every subordinate eligible regardless of performance, severing promotion from performance entirely. Intermediate values $1 < k < n$ preserve a performance requirement while still leaving the superior some latitude over who, among

the qualified, actually receives the prize.

This threshold is the model's central political-design variable. A narrow eligible pool disciplines subordinates through pure competition; a broad one leaves the superior room to reward loyalty instead. The trade-off between the effort a competition elicits and the political control it affords is the mechanism the remainder of this section formalizes.

Stochastic Favoritism. Favoritism in this model cannot simply override the competition: a superior who wants to reward loyalty can act on it only within whichever pool of top performers she has already committed to considering, or her professed reliance on performance becomes transparently hollow. Concretely, the final recipient is selected from the top- k eligible set as follows. If that set contains at least one loyal subordinate, the superior selects uniformly among the loyal subordinates in it: if exactly $q \in \{1, \dots, \min(m, k)\}$ loyal subordinates rank among the top k , each receives the prize with conditional probability $1/q$, and every eligible disloyal subordinate receives it with probability zero. If the top- k set contains no loyal subordinate, the superior selects uniformly among the k eligible disloyal subordinates, so each receives the prize with conditional probability $1/k$.

Favoritism is therefore conditional on performance rather than independent of it. A loyal subordinate cannot receive the prize without first entering the top- k set, and a disloyal subordinate is not mechanically excluded from promotion: a disloyal recipient prevails whenever every eligible subordinate happens to be disloyal. Because the composition of the top- k set depends probabilistically on subordinates' effort choices, the advantage loyalty confers is itself stochastic. Formally, let $P_{li}(x_{li}, \mathbf{x}_{-i}; k)$ and $P_{dj}(x_{dj}, \mathbf{x}_{-j}; k)$ denote the probability that loyal subordinate i and disloyal subordinate j , respectively, receive the prize, and let $P_L(\mathbf{x}; k) \equiv \sum_{i=1}^m P_{li}(\mathbf{x}; k)$ and $P_D(\mathbf{x}; k) \equiv \sum_{j=1}^{n-m} P_{dj}(\mathbf{x}; k)$ denote the resulting probabilities that the prize goes to a loyal or a disloyal subordinate; since exactly one prize is awarded, $P_L(\mathbf{x}; k) + P_D(\mathbf{x}; k) = 1$.

Timing

The sequence of the game is as follows. First, the organizational and institutional parameters (n, m, V, α) are realized and commonly observed. Second, the superior chooses the eligibility threshold k . Third, subordinates simultaneously choose effort, which determines the top- k eligible set through relative performance. Finally, the superior awards the prize according to the stochastic-favoritism rule above.

We solve the game by backward induction: for each k , we first characterize a type-symmetric Nash equilibrium of the subordinate effort subgame, and then determine the superior's optimal eligibility rule given these equilibrium responses.

The Superior

The superior's problem is to set k so as to balance two things she cares about: how much the organization produces, and whether power ultimately rests with someone she trusts. Having anticipated subordinates' equilibrium responses to any threshold k —write $x_l^*(k; m)$ and $x_d^*(k; m)$ for the resulting type-symmetric effort levels, $E^*(k; m) \equiv mx_l^*(k; m) + (n - m)x_d^*(k; m)$ for aggregate output, and $P_L^*(k; m)$, $P_D^*(k; m)$ for the resulting equilibrium probabilities that the prize goes to a loyal or disloyal subordinate—the superior's payoff from choosing k is

$$U_S(k; \alpha, m) = E^*(k; m) + \alpha [P_L^*(k; m) - P_D^*(k; m)] = E^*(k; m) + \alpha [1 - 2P_D^*(k; m)], \quad (1)$$

where the second equality uses $P_L^*(k; m) + P_D^*(k; m) = 1$. The superior then chooses an eligibility rule

$$k^*(\alpha, m) \in \arg \max_{k \in \mathcal{K}} U_S(k; \alpha, m).$$

The first term in (1) is the superior's productive stake in the outcome; the second is its expected authority value, capturing how much the eventual promotion reinforces or erodes her political control.

Subordinates

Subordinates, for their part, face an ordinary career-competition problem. After observing k , each simultaneously chooses effort $x \geq 0$ to maximize the value of the prize, $V > 0$, net of a quadratic effort cost, taking the resulting promotion probability as given: loyal subordinate $i \in \{1, \dots, m\}$ solves $\max_{x_{li} \geq 0} \{P_{li}(x_{li}, \mathbf{x}_{-i}; k)V - x_{li}^2/2\}$, and disloyal subordinate $j \in \{1, \dots, n - m\}$ solves the identical problem with P_{dj} in place of P_{li} . The two types differ only in the promotion probabilities the superior's rule assigns them—not in ability, effort costs, or how much they value the prize—so any equilibrium gap between loyal and disloyal effort is a product of the promotion rule itself, not of an assumed difference in talent or ambition between political insiders and outsiders.

Equilibrium Effort and Organizational Output

We now characterize the subordinate effort subgame for a fixed eligibility threshold k : first the probability that the prize is ultimately awarded to a loyal or disloyal subordinate, then equilibrium effort, and finally the resulting aggregate organizational output.

Winning Probabilities

Under stochastic favoritism, a disloyal subordinate can win only if every member of the top- k eligible set happens to be disloyal, and whether that is even possible depends on how many disloyal subordinates the organization has. If $n - m \geq k$, there are enough disloyal subordinates to fill the entire top- k set, so a disloyal subordinate wins with positive probability whenever disloyal subordinates exert positive effort. If $n - m < k$, by contrast, every feasible top- k set must contain at least one loyal subordinate, and because the superior selects a loyal subordinate whenever one is eligible, disloyal subordinates are then shut out of the prize entirely.

Writing x_l and x_d for the type-symmetric effort levels, the type-level probability that the prize is awarded to a disloyal subordinate is

$$P_D(x_l, x_d; k) = \begin{cases} \frac{(n-m) \binom{n-m-1}{k-1} x_d}{\binom{n-1}{k-1} [m x_l + (n-m) x_d]}, & n - m \geq k, \\ 0, & n - m < k, \end{cases} \quad (2)$$

with $P_L(x_l, x_d; k) = 1 - P_D(x_l, x_d; k)$. Equation (2) captures the central political asymmetry the promotion rule creates: disloyal subordinates keep a positive chance at the prize only so long as it remains feasible for them to occupy every position in the eligible set. Once k exceeds $n - m$, that possibility disappears, and the superior's preference for loyalty forecloses it entirely.

Equilibrium Effort

For each k , we focus on a type-symmetric Nash equilibrium of the subordinate effort subgame.¹ When $n - m \geq k$, both types can win, and combining their first-order conditions yields an equilibrium relationship,

$$x_l^*(k; m) = B(k, m, n) x_d^*(k; m), \quad B(k, m, n) \geq 1, \quad (3)$$

that any interior type-symmetric equilibrium must satisfy; the expression for $B(k, m, n)$ and the underlying first-order conditions are in Appendix . Without presenting the closed-form solutions, we can already characterize how favoritism orders effort across types.

Lemma 1 (Equilibrium Effort under Stochastic Favoritism). *Suppose $1 \leq m \leq n - 1$. For every $k \in \{1, \dots, n\}$, a type-symmetric equilibrium exists, and loyal subordinates*

¹A type-symmetric equilibrium requires symmetry within, but not across, subordinate types: $x_{l_i}^*(k; m) = x_l^*(k; m)$ for every $i = 1, \dots, m$ and $x_{d_j}^*(k; m) = x_d^*(k; m)$ for every $j = 1, \dots, n - m$. Thus $x_l^*(k; m)$ need not equal $x_d^*(k; m)$.

exert weakly more effort than disloyal subordinates:

$$x_l^*(k; m) \geq x_d^*(k; m).$$

More specifically:

1. *If $k = 1$, then $x_l^*(1; m) = x_d^*(1; m) > 0$.*
2. *If $1 < k \leq n - m$, then $x_l^*(k; m) > x_d^*(k; m) > 0$.*
3. *If $n - m < k < n$, then $x_l^*(k; m) > 0 = x_d^*(k; m)$.*
4. *If $k = n$, then $x_l^*(n; m) = x_d^*(n; m) = 0$.*

The four cases trace out exactly how much room stochastic favoritism has to operate. At $k = 1$, only the single top performer is eligible, so the superior has no discretion left to exercise once performance is realized: loyal and disloyal subordinates face identical odds and exert identical effort. Once $1 < k \leq n - m$, favoritism can operate within the eligible set—a loyal subordinate who makes the cut is prioritized in the final allocation, which raises the marginal return to loyalist effort above that of disloyal effort, though disloyal effort remains positive because an all-disloyal eligible set is still possible. Once k exceeds $n - m$, every feasible eligible set necessarily contains a loyalist, so a disloyal subordinate has no path to the prize and rationally stops competing altogether, even as loyal subordinates keep exerting effort to win priority among themselves. Finally, at $k = n$ eligibility is universal and effort no longer affects anything, so both types optimally exert zero effort.

Aggregate Organizational Output

Aggregate organizational output, $E^*(k; m) = mx_l^*(k; m) + (n - m)x_d^*(k; m)$ as defined above, is the total effort exerted by all subordinates. Although stochastic favoritism redistributes effort between loyal and disloyal subordinates, that redistribution does not by itself change the total amount of effort the competition elicits, once the eligibility rule is held fixed.

Proposition 1 (Aggregate Effort Conditional on the Eligibility Rule). *For any fixed $k \in \{1, \dots, n\}$, equilibrium aggregate effort is*

$$E^*(k; m) = \sqrt{V \frac{n - k}{k}}. \quad (4)$$

Conditional on k , aggregate effort is independent of authority capital m and of the superior's reliance on loyalty α . It strictly decreases in k and equals zero at $k = n$: a more restrictive eligibility rule induces greater aggregate effort.

Proposition 1 separates the incentive effect of the eligibility rule from the distributional effect of favoritism. A larger k broadens the eligible set and weakens the link between relative performance and promotion, so subordinates face a lower marginal return to effort and aggregate output falls; at the limit $k = n$, promotion is entirely decoupled from performance and aggregate effort vanishes. Authority capital m , by contrast, changes only how a given amount of effort is *distributed* across loyal and disloyal subordinates for a fixed k —loyal subordinates exert more, disloyal subordinates less, and the two changes exactly offset each other in the aggregate.

This conditional invariance does not make m or α irrelevant to organizational output. Both remain exogenous in the baseline model, but they shape the superior's choice of $k^*(\alpha, m)$ itself—the eligibility rule she actually selects, defined above—and it is through that choice, rather than through any direct effect at a fixed k , that authority capital and reliance on loyalty end up shaping realized output $\hat{E}(\alpha, m) \equiv E^*(k^*(\alpha, m))$. The remainder of this section characterizes that institutional-choice channel. The proof of Proposition 1 is in Appendix .

Figure 1 illustrates the patterns in Lemma 1 and Proposition 1: loyal effort remains positive over a wider range of eligibility rules than disloyal effort, because relative performance keeps mattering among loyalists even after disloyal subordinates have been shut out; and aggregate effort declines as k grows, but at any fixed k is invariant to authority capital m , which reshuffles the composition of effort without changing its total.

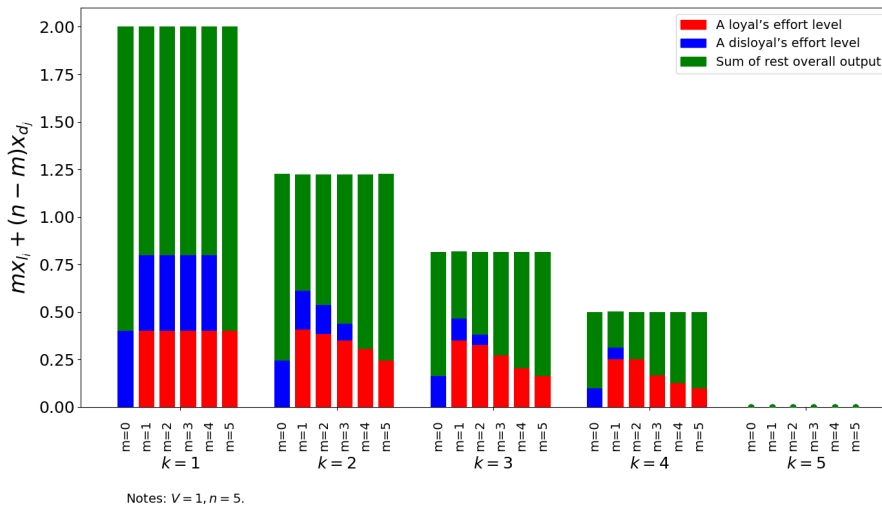


Figure 1 Equilibrium Effort and Aggregate Output Conditional on the Eligibility Rule

The Optimal Choice of Competition

Having characterized the subordinate effort subgame, we turn to the superior's choice of k itself. A smaller k buys more intense, performance-based competition; a larger k buys more room for loyalty to shape the outcome. For notational convenience, let

$A(k; m) \equiv 1 - 2P_D^*(k; m)$ denote the expected authority value of the promotion outcome under rule k , and, using (1) together with the fact that aggregate effort $E^*(k) = \sqrt{V(n-k)/k}$ does not depend on m for fixed k , write the superior's problem as choosing k to maximize $E^*(k) + \alpha A(k; m)$ over $\mathcal{K} = \{1, \dots, n\}$. The superior's institutional choice is therefore a straightforward trade-off: a smaller eligible set raises aggregate output, a larger one raises the scope for using loyalty to determine the final recipient.

Pairwise Institutional Comparisons

Because k is discrete, this trade-off resolves rule by rule. Consider two rules $k' > k$. Moving from k to k' costs the superior $E^*(k) - E^*(k') > 0$ in aggregate output, but it can gain her authority value $\alpha[A(k'; m) - A(k; m)]$ whenever the broader pool raises the odds that a loyalist ultimately wins. Whenever $A(k'; m) > A(k; m)$, there is accordingly a reliance-on-loyalty threshold

$$\hat{\alpha}_{k,k'}(m) \equiv \frac{E^*(k) - E^*(k')}{A(k'; m) - A(k; m)} \quad (5)$$

above which the superior prefers the broader rule k' to k , and below which she prefers k . Because her overall optimum is the upper envelope of these pairwise comparisons across all $k \in \mathcal{K}$, some intermediate rules may never be optimal for any α —the optimal rule can jump directly from, say, $k = 1$ to $k = 3$ without $k = 2$ ever being chosen. Except at the finite set of α where two rules tie, the optimal rule $k^*(\alpha, m)$ is unique.

Lemma 2 (Benchmark without Reliance on Loyalty). *When $\alpha = 0$, the superior chooses the strict tournament, $k^*(0, m) = 1$ for every m .*

With no political stake in who wins— $\alpha = 0$ —the superior maximizes aggregate output alone, and since $E^*(k)$ is strictly decreasing in k , the strict tournament is her unique optimum. This benchmark isolates the pure productivity motive before the two political forces of interest, reliance on loyalty and authority capital, are allowed to operate.

Reliance on Loyalty and the Optimal Rule

Expanding the eligible set raises the probability that loyalty can shape the final outcome, so the expected authority value to a broader rule is weakly increasing in α .

Proposition 2 (Reliance on Loyalty and Competition). *For any given authority capital m , the optimal eligibility threshold $k^*(\alpha, m)$ is weakly increasing in the superior's reliance on loyalty α : $\alpha' \geq \alpha \implies k^*(\alpha', m) \geq k^*(\alpha, m)$. Equivalently, weaker legal institutions induce weakly less intense performance-based competition.*

The logic is monotonic and intuitive: for any two rules $k' > k$, the relative payoff of the broader rule rises with α , so once the superior prefers k' at some level of reliance on

loyalty, a further increase in α can never reverse that preference. Politically, stronger legal institutions reduce the superior's need to lean on loyal subordinates to secure compliance, so she places relatively more weight on aggregate output and sustains a tighter, more performance-based rule. As legal institutions weaken, loyalty becomes correspondingly more valuable, and a broader eligibility rule becomes increasingly attractive.

Authority Capital and Institutional Choice

Authority capital's effect is more nuanced, because it operates only through how it changes the marginal authority gain from broadening the eligible set. For any $k' > k$, define this gain as

$$G_{k,k'}(m) \equiv A(k'; m) - A(k; m). \quad (6)$$

Because aggregate output at a fixed k does not depend on m , authority capital shapes the superior's institutional choice only by moving $G_{k,k'}(m)$ —and depending on how it moves, three distinct institutional logics can arise.

Proposition 3 (Authority Capital and Institutional Regimes). *The effect of authority capital on the optimal eligibility rule can be characterized as follows.*

1. **Authority-capital neutrality.** *In a neighborhood of $\alpha = 0$, $k^*(\alpha, m) = 1$ for every feasible m : changes in authority capital leave the superior's optimal rule unchanged.*
2. **Discretion activation.** *For some $k' > k$, an increase in authority capital activates promotion discretion when the additional loyalist raises the marginal authority gain from broadening, $G_{k,k'}(m+1) > G_{k,k'}(m)$, and the productivity cost of the move falls between the two resulting thresholds, $\alpha G_{k,k'}(m) < E^*(k) - E^*(k') \leq \alpha G_{k,k'}(m+1)$. Under these conditions, adding a loyalist causes the superior to switch from the stricter rule k to the broader rule k' .*
3. **Authority–discretion substitution.** *If $G_{k,k'}(m)$ is weakly decreasing in m for every $k' > k$, then $k^*(\alpha, m+1) \leq k^*(\alpha, m)$: greater authority capital weakly substitutes for promotion discretion and supports a more restrictive rule.*

Figure 2 summarizes these three regions schematically; because α measures reliance on loyalty, low α corresponds to strong legal institutions and high α to weak ones.

The three regions correspond to three distinct political situations. Where legal institutions are strong and reliance on loyalty is correspondingly low, the productive advantage of the strict tournament simply dominates the authority benefit of any broader

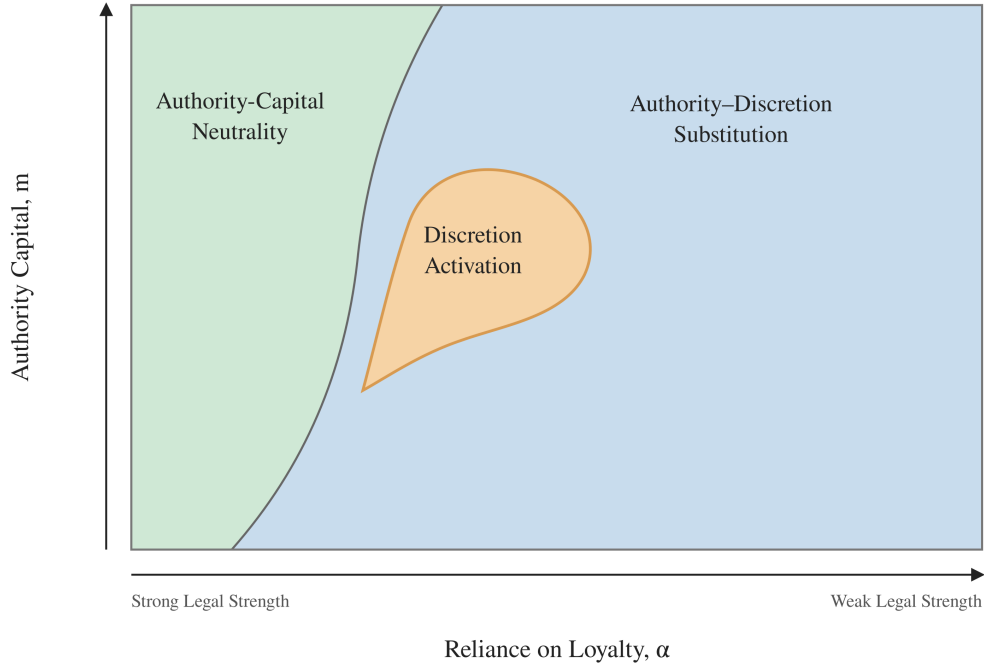
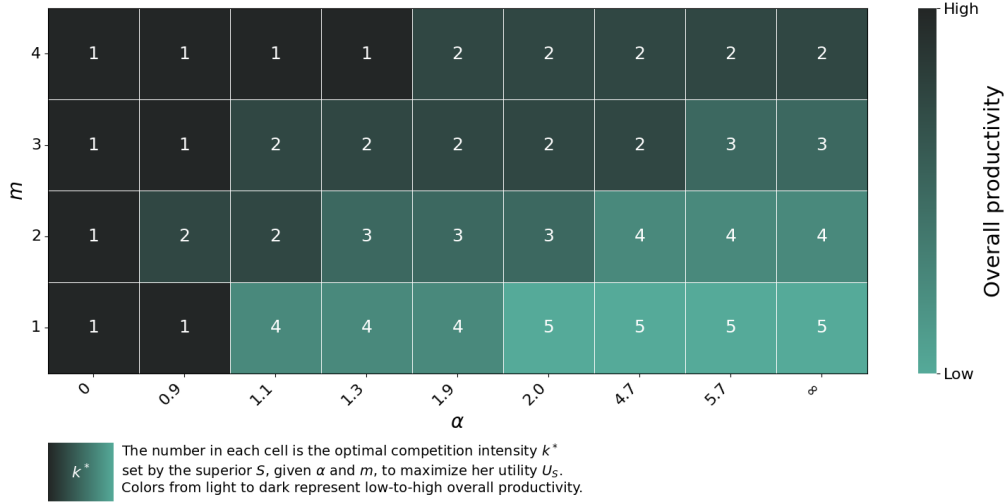


Figure 2 AUTHORITY CAPITAL, PROMOTION DISCRETION, AND INSTITUTIONAL REGIMES

rule: authority capital may still affect who ends up winning, but it does not change the superior's optimal k . Where the loyalty network starts out thin, adding a single loyalist can activate promotion discretion for the first time—if the resulting authority gain crosses the threshold needed to offset the output it costs, the superior broadens the pool; because this requires a specific crossing of the two margins, the activation effect is necessarily localized rather than pervasive. And where the superior already commands a substantial loyal following, an additional loyalist instead substitutes for discretion: she can restore a tighter, more productive rule precisely because political control no longer depends on keeping the pool broad. The relationship between authority capital and competition intensity is therefore not globally monotonic, but outside the narrow activation region, more authority capital either leaves the optimal rule unchanged or makes it weakly more restrictive.

Figure 3 illustrates the global solution numerically for $n = 5$ and $V = 1$; darker cells indicate a smaller optimal $k^*(\alpha, m)$ and therefore higher output.

The heat map illustrates all three regimes at once: holding m fixed, the optimal rule is weakly increasing in α , consistent with Proposition 2; across most of the parameter space, increasing authority capital leaves k^* unchanged or lowers it, the authority–discretion substitution effect; and a localized pocket exists in which increasing m instead raises k^* —not a numerical artifact, but the discretion-activation region, where the extra loyalist shifts the global upper envelope of the superior's payoff functions enough to make the broader rule newly attractive. Figure 3 reports the exact global solution



Notes: $V = 1$, $n = 5$.

Figure 3 OPTIMAL ELIGIBILITY RULES ACROSS RELIANCE ON LOYALTY AND AUTHORITY CAPITAL

for this numerical example, while Figure 2 distinguishes the underlying political logic operating in each region of the parameter space.

Corollary 1 (Induced Productivity Effects). *Because aggregate equilibrium output is strictly decreasing in k , the institutional comparative statics above imply the following productivity effects:*

1. *Aggregate output is weakly decreasing in the superior's reliance on loyalty α .*
2. *In the authority-capital-neutrality region, changes in m leave aggregate output unchanged.*
3. *In the discretion-activation region, an increase in m may reduce aggregate output by inducing a broader eligibility rule.*
4. *In the authority-discretion-substitution region, an increase in m weakly raises aggregate output by supporting a more restrictive eligibility rule.*

The upshot is that weak legal institutions alone do not determine organizational productivity. What matters is how reliance on loyalty interacts with the superior's existing stock of authority capital: a thin loyalty network can push her to relax performance-based competition, while a sufficiently dense one lets her restore a stricter rule instead.

Political Inequality

Recall that aggregate equilibrium output is

$$E^*(k) = mx_l^*(k; m) + (n - m)x_d^*(k; m) = \sqrt{V \frac{n - k}{k}}. \quad (7)$$

For a given eligibility rule k , aggregate output is independent of authority capital m . By contrast, politically induced effort inequality is generally determined jointly by the eligibility threshold and the composition of the subordinate population:

$$I^*(k; m) = x_l^*(k; m) - x_d^*(k; m).$$

This separation implies that aggregate productivity and the distribution of effort across subordinate types are distinct organizational outcomes.

Proposition 4 (Separation of Productivity and Political Inequality). *For fixed n and V , aggregate equilibrium output is determined by the eligibility threshold k , whereas politically induced effort inequality is determined jointly by k and authority capital m . Consequently:*

1. *Political inequality can arise at every interior productivity level generated by an intermediate eligibility rule $k \in \{2, \dots, n - 1\}$.*
2. *Holding k fixed, variations in authority capital can alter the equilibrium effort gap without changing aggregate output.*
3. *There is no globally monotonic relationship between aggregate productivity and politically induced effort inequality.*

The first part of the proposition follows because every intermediate eligibility rule produces both a positive effort gap and an interior level of aggregate output:

$$I^*(k; m) > 0 \quad \text{and} \quad 0 < E^*(k) < E^*(1) \quad \text{for every } 1 < k < n.$$

This does not imply that political inequality can occur at literally every possible level of productivity. The two boundary productivity levels are excluded. The strict tournament $k = 1$ produces maximum aggregate output without an effort gap, whereas unrestricted eligibility $k = n$ produces zero aggregate output and no effort gap. Rather, political inequality can arise at every interior productivity level generated by the model.

The second part establishes a stronger separation between productivity and inequality. For any two feasible values of authority capital m and m' ,

$$E^*(k; m) = E^*(k; m')$$

when k is held fixed. However, because individual equilibrium effort levels generally depend on the composition of the subordinate population, it is possible that

$$I^*(k; m) \neq I^*(k; m').$$

Organizations with the same eligibility threshold—and therefore the same aggregate output—may consequently exhibit different effort gaps between loyal and disloyal subordinates.

Finally, the relationship between productivity and political inequality cannot be globally monotonic. Aggregate output is strictly decreasing in the eligibility threshold:

$$E^*(1) > E^*(2) > \dots > E^*(n) = 0.$$

Political inequality, by contrast, is zero at both endpoints but strictly positive under every intermediate rule:

$$I^*(1; m) = I^*(n; m) = 0,$$

and

$$I^*(k; m) > 0 \quad \text{for every } 1 < k < n.$$

Thus, high productivity can coexist with no political inequality under a strict tournament, while lower interior productivity levels can coexist with positive political inequality. At the opposite extreme, the complete collapse of performance incentives produces both zero output and zero effort inequality. Productivity and political inequality therefore need not move together.

Political Inequality in the Final Equilibrium. The results above characterize effort inequality conditional on an eligibility rule k . To characterize political inequality in the final equilibrium, substitute the superior's optimal eligibility rule $k^*(\alpha, m)$ into the effort-gap function:

$$\mathcal{I}^*(\alpha, m) \equiv I^*(k^*(\alpha, m); m). \tag{8}$$

Reliance on loyalty α affects political inequality indirectly through the superior's choice of the eligibility threshold $k^*(\alpha, m)$. Authority capital m , by contrast, affects political inequality through two channels. First, it directly changes the relative effort incentives of loyal and disloyal subordinates, conditional on k . Second, it indirectly affects inequality by changing the superior's optimal institutional choice.

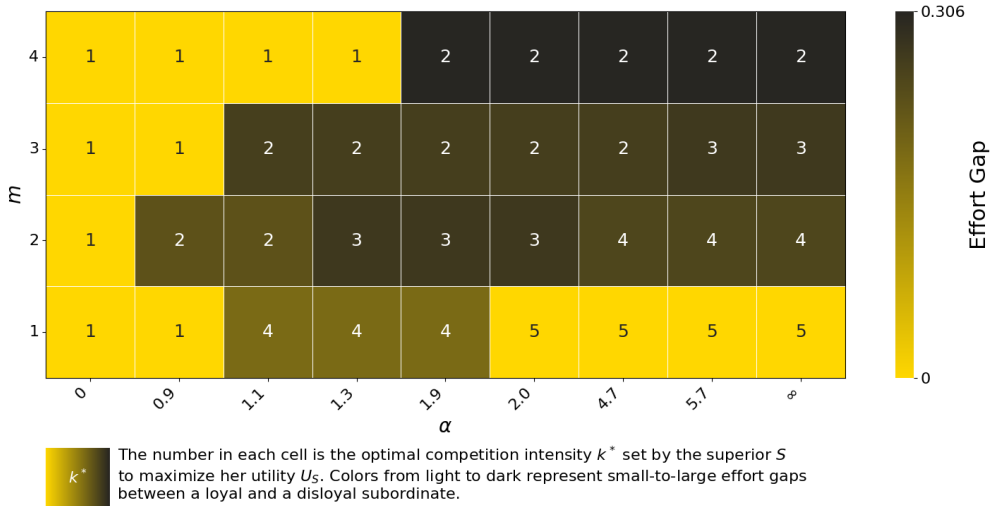
Corollary 2 (Political Inequality in the Final Equilibrium). *Final-equilibrium political inequality is strictly positive if and only if the superior optimally selects an intermediate eligibility rule:*

$$\mathcal{I}^*(\alpha, m) > 0 \iff 1 < k^*(\alpha, m) < n. \quad (9)$$

Political inequality is therefore endogenous to the superior's competition design. It is not generated by exogenous differences in productive ability or preferences between subordinate types. Instead, it arises because the superior adopts an institutional arrangement in which performance determines eligibility while loyalty determines priority within the eligible set.

Because $k^*(\alpha, m)$ changes discretely across the parameter space, final-equilibrium political inequality need not vary monotonically with either reliance on loyalty or authority capital. Changes in α or m may move the organization into or out of the intermediate institutional region in which performance competition and loyalty-based favoritism operate jointly.

Figure 4 illustrates the relationship among reliance on loyalty α , authority capital m , the superior's optimal eligibility rule $k^*(\alpha, m)$, and final-equilibrium political inequality for $n = 5$ and $V = 1$.



Notes: $V = 1$, $n = 5$.

Figure 4 OPTIMAL ELIGIBILITY RULES AND POLITICALLY INDUCED EFFORT INEQUALITY

Consistent with Corollary A.3, effort gaps appear only in parameter regions in which the superior selects an intermediate eligibility rule. When $k^* = 1$, promotion is determined exclusively by relative performance and loyal and disloyal subordinates exert identical effort. When $k^* = n$, promotion is entirely independent of performance and both types exert zero effort.

The numerical example also illustrates the separation between productivity and political inequality. Aggregate output depends only on the selected eligibility threshold, whereas the effort gap depends additionally on authority capital. Organizations with

the same aggregate productivity may therefore exhibit different degrees of politically induced effort inequality.

The figure should be interpreted as an illustration of the general analytical results rather than as their proof. The analytical results apply to any organizational size n , prize value V , and composition m satisfying the assumptions of the model, whereas the figure reports the final-equilibrium outcomes for the specific case $n = 5$ and $V = 1$.

Endogenous Authority Capital and Institutional Adjustment

So far the superior has taken both her legal environment and her loyal following as inherited facts of political life. Neither actually is. A loyal network is built, subordinate by subordinate, and legal institutions are not immutable either—they can be reinforced, quietly hollowed out, or openly dismantled. This extension lets the superior do both, over two periods rather than one. The two technologies of control are not perfect substitutes: authority capital buys compliance through personal loyalty, while formal legal enforcement buys it by making commands predictable and impersonally enforceable, and carries a private benefit to the superior quite apart from whatever it does to the promotion contest. Which technology she ends up favoring, and whether she reinforces or erodes the rule of law along the way, turns out to depend entirely on how much authority capital she has already built.

The organization inherits a state (m_0, α_0) at the start of period 0, exactly as in Section , and aggregate effort $E^*(k)$ and the authority value $A(k; m)$ retain their meaning from Proposition 1. What is new is a private compliance benefit that formal law confers on the superior directly: legal enforcement of effectiveness $\ell(\alpha)$, decreasing and convex in α and vanishing as reliance on authority capital grows without bound (our benchmark specification is $\ell(\alpha) = 1/(1 + \alpha)$), so that her flow payoff is

$$u(k, m, \alpha) = E^*(k) + \alpha A(k; m) + \beta \ell(\alpha), \quad \beta > 0, \quad (10)$$

with $\widetilde{W}(m, \alpha) \equiv \max_k u(k, m, \alpha)$ the value once competition is optimized out. Because the compliance term does not depend on k , it never distorts the superior's choice of competition rule at a given (m, α) ; it only reshapes which institutional environment she would rather inhabit.

Two periods separate her from that environment, and she moves through them in order. In period 0 she still competes under the inherited rules, banking $\widetilde{W}(m_0, \alpha_0)$; only afterward does she decide whether to expand or dismantle her loyal following—there is no depreciation, so whatever she builds or tears down simply stays that way. Only once her new authority capital is in hand does she confront the legal environment itself, choosing whether to strengthen, preserve, or weaken formal enforcement, together with a fresh competition rule for the period ahead.

Condition 1 (Costs of Adjustment). *Expanding the loyal network is cheaper the weaker is α_0 ; dismantling it entails a separate variable cost that does not depend on α_0 . Strengthening the law entails a fixed and a variable cost that both rise with authority capital; weakening it entails a fixed and a variable cost that both fall with authority capital.*

In short, patronage is cheap to build only where courts are already weak, and a large loyal coalition makes defending the law more costly and abandoning it cheaper—the exact functional forms are given in the Appendix. Discounting the second period at $\delta \in (0, 1]$, the superior’s total payoff is simply the sum of the two periods’ flows, net of whatever each adjustment costs her to make.

Solving backward, fix the authority stock m_1 she carries into period 1 and the inherited α_0 . Three paths are open to her: strengthen the law, preserve it, or weaken it further. Call the resulting continuation values $S(m_1, \alpha_0)$, $Q(m_1, \alpha_0)$, and $D(m_1, \alpha_0)$ —each the best she can do along that path once competition is re-optimized, net of cost. She takes the largest of the three, with ties resolved toward the status quo. The three results below characterize this choice, first at the margin and then globally; all proofs and exact functional forms are collected in the Appendix.

Proposition 5 (Competing sources of compliance). *Fix (k, m) with $A(k; m) > 0$. The superior’s marginal incentive to rely on authority capital rather than formal law changes sign at*

$$\hat{\alpha}(k, m) = \max \left\{ 0, \sqrt{\beta/A(k; m)} - 1 \right\};$$

she prefers strengthening the law when $\alpha < \hat{\alpha}(k, m)$, and weakening it when $\alpha > \hat{\alpha}(k, m)$. If $A(k; m) \leq 0$, she prefers weakening it at every $\alpha \geq 0$.

At the margin, then, loyalty and law substitute for one another around a single crossing point—but that only describes the direction she would drift if institutional change were costless and continuous. Fixed costs, together with competition itself re-adjusting along the way, mean the global decision can look quite different.

Proposition 6 (Institutional inaction). *The status quo survives whenever neither direction of reform offers a gain, net of its variable cost, large enough to cover its fixed cost; otherwise the superior adopts whichever direction offers the larger net gain. (Formal thresholds and proof are given in the Appendix.)*

Fixed costs, in other words, can freeze the law in place well past the point where the superior would already prefer to change it if change were free—institutions are stickier than the incentives to alter them.

Proposition 7 (Authority capital and institutional weakening). *As the superior’s authority capital grows, the range of inherited legal environments in which weakening*

the law is optimal weakly expands, in the sense that once weakening becomes optimal at some authority-capital level it remains optimal at every higher level, so a well-defined tipping point separates the two regions. Under the calibration reported in Figure 5, every inherited state additionally falls into exactly one of three ordered regions: strengthen, hold, or weaken, separated by cutoffs that shift with authority capital.

This is the mechanism that ties the two technologies of control together across periods: a superior who has already accumulated a large loyal following finds weakening the law both more attractive and less costly—precisely because that following is what she was accumulating in period 0 in the first place. Anticipating exactly this, she chooses, already in period 0, how far to expand or dismantle her following by weighing its cost against the discounted continuation value it buys her; because authority capital is discrete, this amounts to comparing a finite set of numbers, with no concavity assumption or depreciation term required.

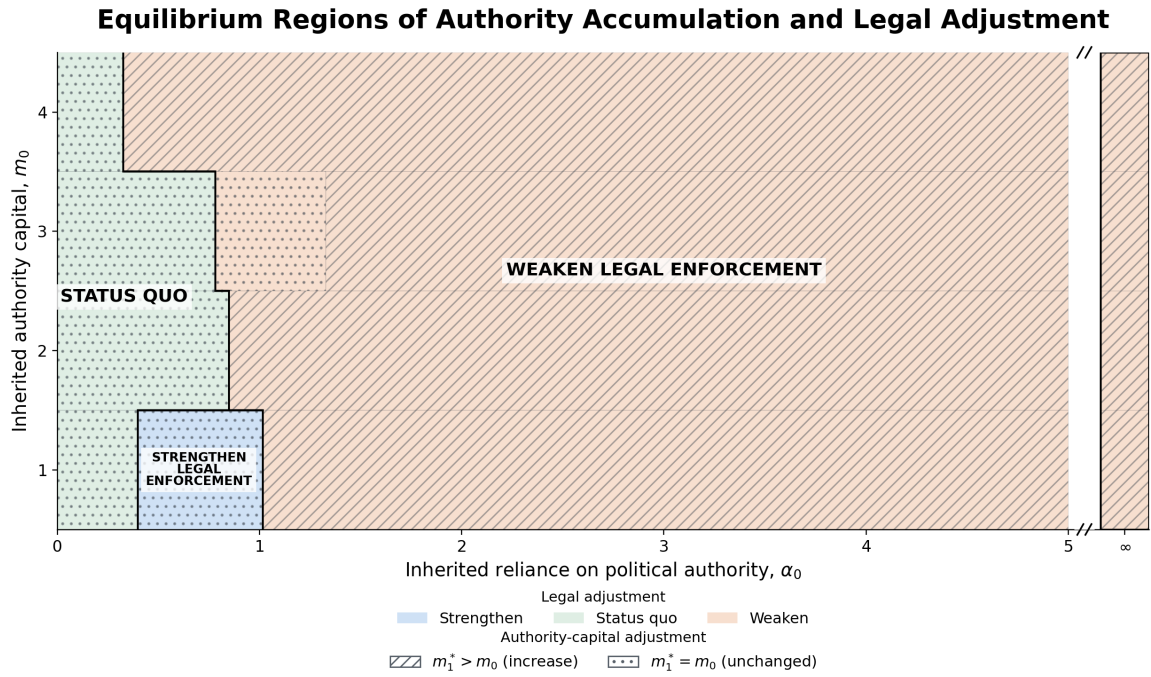
A numerical calibration, using the exact baseline expressions for $E^*(k)$ and $A(k; m)$ together with standard parameter values, makes the mechanism concrete. For every inherited state we enumerate the full choice set—every competition rule, every institutional adjustment in both directions, and the status quo—and select the global optimum; the resulting equilibrium map, with the period-0 authority-capital choice already embedded, is shown in Figure 5.

Figure 5 overlays the sign of the endogenous authority-capital adjustment—right-diagonal hatching where the superior expands her following, dots where she leaves it unchanged. The pattern is the paper’s backsliding story rendered as a single mechanism: the more authority capital a superior begins with, the wider the range of legal environments in which she chooses to weaken the law further rather than defend it, and under this calibration no inherited state ever induces a reduction in authority capital. The calibration is illustrative rather than estimated; the object of substantive interest is the qualitative comparison in the three propositions above, and alternative parameterizations belong in a sensitivity analysis rather than a claim about magnitudes.

Implications: Productivity, Inequality, and Institutional Stability

Both halves of the model point to the same underlying claim: authority capital and legal enforcement are substitutes for securing compliance, and how a superior balances them—not her regime type, her technology, or her culture—is what ultimately governs productivity, inequality, and the durability of the institutions she operates under.

The baseline model’s first implication concerns productivity. Differences in out-



Global numerical solution under the reported benchmark calibration. The horizontal axis is linear on $\alpha_0 \in [0, 5]$; the narrow broken-axis strip shows the verified $\alpha_0 \rightarrow \infty$ outcome.

Figure 5Equilibrium regions of authority accumulation and legal adjustment. Since m_0 and m_1 are discrete, the numerical boundaries are step-shaped rather than smooth. Colors identify legal adjustment; hatching identifies the equilibrium change in authority capital.

put need not originate from differences in technology, culture, or regime type per se; they can arise endogenously from how internal competition is designed under authority constraints. When a superior relies heavily on loyalty-based authority, she may strategically relax competition intensity to preserve authority rents, even at the cost of weaker effort incentives—so otherwise similar organizations can exhibit sharply different productivity simply because their authority capital and legal environment support different competition rules. This perspective helps reconcile a long-standing disagreement over regime type and economic performance: rather than democracy or autocracy per se, both can sustain either high or low productivity, depending on how authority concerns interact with internal competition. Weak legal environments do not mechanically imply low productivity—when authority capital is sufficiently strong, a superior can still enforce intense competition and sustain high output—and, conversely, even formally democratic systems can see productivity decline if authority concerns lead them to relax competition. The same logic operates inside firms: leaders with similar technologies but different authority capital or reliance on loyalty will choose different promotion rules, so competition intensity emerges as a political rather than a purely technological or market-driven parameter.

The second implication concerns inequality. Effort gaps between favored and disfavored subordinates are not a direct consequence of low productivity or weak competition; they are a political byproduct of authority-preserving competition design, which is why severe effort inequality can coexist with high productivity under intermediate competition rules, while minimal inequality can accompany both the most intense and the weakest forms of competition—inequality and aggregate performance are jointly shaped by authority maintenance but not mechanically linked to one another. The same logic extends beyond organizations to state-level governance: when political control is maintained through identity-based favoritism, incentive structures become uneven across groups, and economic inequality emerges not primarily from differential market returns but from systematically induced differences in effort and participation—a downstream consequence of politically generated effort inequality that complements existing political science accounts of how social cleavages translate into persistent economic disparities (Lipset and Rokkan 1967; Alesina and Glaeser 2004; Padgett and McLean 2006).

These are comparative statics for a given stock of authority capital, however, and the natural next question is where that stock comes from and whether it is stable. The two-period extension answers both. Hierarchical organizations endogenously generate incentives to accumulate political authority—not as a distortion of otherwise efficient competition, but as a direct consequence of how promotion-based competition is designed: whenever a superior can influence the composition of her loyal following, she strictly prefers to expand it, since a denser loyalty network enlarges the range of com-

petition rules that can sustain high performance without jeopardizing political control. This preference holds across legal regimes and applies wherever hierarchical structure and internal competition coexist, from firms and bureaucracies to states.

That same accumulation, however, is what eventually makes strong legal enforcement difficult to sustain. When legal enforcement is strong and authority capital is still small, weakening the law is unattractive to the superior; but this region is bounded by a tipping point, and once authority capital grows past it, her incentives reverse—weakening the law becomes privately optimal even though strong enforcement remains socially efficient. Partial legal reforms are consequently fragile: unless a reform pushes legal strength well past the tipping point, or directly constrains how much authority capital a superior can accumulate, private incentives eventually reassert themselves and legal enforcement erodes on its own. This gives a structural account of institutional backsliding: a high-enforcement, low-inequality equilibrium is not self-enforcing if superiors retain discretion over authority accumulation, since crossing the tipping point gives them a private incentive to weaken legal constraints even while fully aware that strong rule of law is socially beneficial. Backsliding, in this light, need not reflect cultural regression or a lack of modernization; it can arise endogenously from the interaction between hierarchical authority, discretionary power, and competition design—which is also why sustaining a stable, high-rule-of-law equilibrium requires institutions that limit both the expansion of authority capital and the discretionary erosion of legal enforcement, rather than institutions that address only one.

Taken together, these results suggest that institutional reforms aimed solely at strengthening formal rules or intensifying competition will have limited, and possibly temporary, effects unless they also confront the authority structures that govern how that competition is actually implemented.

Conclusion

A recurring concern in political science is how compliance and order are sustained in organizations, including bureaucracies, parties, firms, and states, when formal legal enforcement is weak, contested, or simply absent. Studies of clientelism, patronage, and informal institutions have long shown that loyalty-based ties often do the work that formal rules cannot. What has been missing is a formal account of how this substitution actually operates: how much authority a superior needs to build outside the law, and what that construction costs in terms of competition, effort, and inequality within the organization.

This paper offers such an account. It argues that superiors respond to weak or uncertain legal enforcement by building authority capital within their hierarchies, cul-

tivated through the ordinary politics of personnel decisions: whom to hire, promote, and reward, and how much loyalty is allowed to compete with performance as a criterion for advancement. A superior who cannot fully depend on courts or formal rules to enforce compliance can instead shape the composition of her own following, trading some measure of competitive intensity for a durable base of support. Authority, on this view, is not simply given by the legal environment; it is something organizations actively build up as capital.

Because this construction is deliberate rather than incidental, it carries consequences well beyond the immediate question of compliance. It shapes how hard subordinates compete, how unequally they are treated, and how that treatment differs across otherwise similar organizations. And because authority capital is something a superior can accumulate over time, its logic extends past any single equilibrium: today's rational response to weak enforcement can become tomorrow's incentive to erode enforcement that is, in fact, working well. Read together with the paper's substantive results, this suggests that legal reform and organizational reform are not independent projects. Strengthening the law without addressing the authority capital organizations build around it is unlikely to produce lasting change.

More broadly, the account offered here suggests that authority capital deserves treatment as a first-order variable in the study of organizations, not as a residual explained away by culture, regime type, or technology. Firms, bureaucracies, and states all face variants of the same underlying problem, and the same logic of authority accumulation appears to operate across them. A natural next step is tracing that logic through specific historical episodes: organizations and states that built authority capital to compensate for weak legal enforcement, and others where that capital later became the source of institutional decay. This framework is well positioned to guide that inquiry.

Appendix

This appendix derives every result whose proof the main text defers here. Section derives the contest success function used throughout. Section proves every Lemma, Proposition, and Corollary stated in Section 3 ("The Model"), including why the optimal competition rule cannot in general be claimed monotone in authority capital, and a proposition showing that legal weakness does not impose an unconditional ordering on productivity once authority capital is allowed to differ across environments. Section proves the results of Section 5 (the two-period extension).

The Berry Multi-Winner Contest Success Function

The winning probabilities used in the main text are an application of Berry's (?) multi-winner contest success function, which this section derives from the single-winner Tullock contest success function (TSF) it generalizes.

Let $x = (x_1, \dots, x_n)$ denote a vector of efforts for n players. A single-winner contest success function assigns each player i a winning probability $Prob^i(x)$ satisfying three standard properties (Skaperdas 1996): (1A) probabilities are nonnegative and sum to one, and $Prob^i(x) > 0$ whenever $x_i > 0$; (2A) $Prob^i(x)$ is increasing in x_i and decreasing in every $x_j, j \neq i$; (3A) $Prob^i(x)$ is invariant to relabeling players. The Tullock success function,

$$Prob^i(x) = \frac{x_i^r}{x_1^r + x_2^r + \dots + x_n^r}, \quad (A1)$$

satisfies all three properties, where r indexes the responsiveness of winning probability to effort (Rai and Sarin 2009).

Berry (1993) extends (A1) to a k -winner contest ($1 < k \leq n$) by summing, over every combination in which player i is one of the k winners, the TSF-weighted share of that outcome. The numerator sums x_i^r across the $\binom{n-1}{k-1}$ combinations of the remaining $k-1$ winners together with the efforts of those co-winners; the denominator sums the same TSF-style expression across all $\binom{n}{k}$ possible winning sets. The resulting probability that player i is among the top k is

$$Prob_k^i(x) = \frac{\binom{n-1}{k-1} x_i^r + \binom{n-2}{k-2} \sum_{j \neq i} x_j^r}{\binom{n-1}{k-1} \sum_{z=1}^n x_z^r}, \quad j, i \in N. \quad (A2)$$

Setting $r = 1$ in (A2) and applying the stochastic-favoritism selection rule of Section 3 (a loyal winner is drawn uniformly from the loyal members of the top- k set whenever one exists, and a disloyal winner only otherwise) yields the individual winning probabilities P_{li}, P_{dj} used throughout the main text and this appendix.

General Proofs for the Baseline Model

Throughout this section let

$$D \equiv n - m$$

denote the number of disloyal subordinates, and let $\mathcal{K} \equiv \{1, \dots, n\}$ be the superior's feasible set of competition rules. Including $k = n$ is necessary because $k = n$ is the case in which eligibility, and hence promotion, is completely independent of performance. Unless otherwise stated, results comparing loyal and disloyal agents presume $1 \leq m \leq n - 1$; the boundary cases $m = 0$ and $m = n$ contain only one type.

Anticipating the main text's Proposition 1, define

$$E^*(k) \equiv \sqrt{\frac{V(n-k)}{k}}, \quad k \in \mathcal{K}, \quad (\text{A3})$$

with $E^*(n) = 0$; this section derives, rather than assumes, that equilibrium aggregate effort equals this expression. For $k \leq D$, also define

$$R_k(m) \equiv \frac{\binom{n-1}{k}}{\binom{D}{k}}, \quad Q_k(m) \equiv k + DR_k(m), \quad (\text{A4})$$

and

$$B_k(m) \equiv \frac{DR_k(m) - D + k}{m}. \quad (\text{A5})$$

$B_k(m)$ is the quantity denoted $B(k, m, n)$ in the main text (eq. (3)), with the dependence on n suppressed since $D \equiv n - m$.

Preliminary binomial identities

We repeatedly use Vandermonde's identity,

$$\sum_r \binom{a}{r} \binom{b}{s-r} = \binom{a+b}{s}, \quad (\text{A6})$$

where a binomial coefficient is understood to be zero whenever its lower index lies outside its admissible range. We also use

$$\binom{a}{b-1} \frac{a-b+1}{b} = \binom{a}{b} \quad (\text{A7})$$

and

$$D \binom{D-1}{k-1} = k \binom{D}{k}. \quad (\text{A8})$$

The symmetric effort equilibrium

Lemma A.1 (Symmetric equilibrium effort). *Suppose $1 \leq m \leq n-1$ and $V > 0$.*

1. *If $k \leq D$, the interior symmetric equilibrium is*

$$x_d^*(k; m) = \frac{E^*(k)}{Q_k(m)}, \quad x_l^*(k; m) = B_k(m) \frac{E^*(k)}{Q_k(m)}. \quad (\text{A9})$$

2. *If $D < k < n$, the symmetric equilibrium is*

$$x_d^*(k; m) = 0, \quad x_l^*(k; m) = \frac{E^*(k)}{m}. \quad (\text{A10})$$

3. If $k = n$, the unique equilibrium is

$$x_d^*(n; m) = x_l^*(n; m) = 0. \quad (\text{A11})$$

Proof. We first consider $k \leq D$. At a type-symmetric effort profile, write the effort of every loyal agent as x_l and that of every disloyal agent as x_d . Differentiating the contest-success functions with respect to an individual agent's own effort and then imposing type symmetry gives

$$\frac{\partial P_{li}}{\partial x_{li}} = \frac{x_l \sum_{q=1}^{\min\{m,k\}} \binom{m-1}{q} \binom{D}{k-q} + \frac{D}{m} x_d \sum_{q=1}^{\min\{m,k\}} \binom{m}{q} \binom{D-1}{k-q}}{\binom{n-1}{k-1} (mx_l + Dx_d)^2}, \quad (\text{A12})$$

$$\frac{\partial P_{dj}}{\partial x_{dj}} = \frac{\frac{m}{D} \binom{D}{k} x_l + \binom{D-1}{k} x_d}{\binom{n-1}{k-1} (mx_l + Dx_d)^2}. \quad (\text{A13})$$

By Vandermonde's identity,

$$\sum_{q=1}^{\min\{m,k\}} \binom{m-1}{q} \binom{D}{k-q} = \binom{n-1}{k} - \binom{D}{k}, \quad (\text{A14})$$

$$\sum_{q=1}^{\min\{m,k\}} \binom{m}{q} \binom{D-1}{k-q} = \binom{n-1}{k} - \binom{D-1}{k}. \quad (\text{A15})$$

The first-order conditions are

$$x_l = V \frac{\partial P_{li}}{\partial x_{li}}, \quad x_d = V \frac{\partial P_{dj}}{\partial x_{dj}}. \quad (\text{A16})$$

Because all coefficients in (A12)–(A13) are nonnegative and at least one coefficient in each expression is positive, an equilibrium in this region has $x_l > 0$ and $x_d > 0$. Let $B = x_l/x_d$. Dividing the loyal first-order condition by the disloyal first-order condition, and using (A14)–(A15), yields the quadratic condition

$$B \left[\frac{m}{D} \binom{D}{k} B + \binom{D-1}{k} \right] = B \left[\binom{n-1}{k} - \binom{D}{k} \right] + \frac{D}{m} \left[\binom{n-1}{k} - \binom{D-1}{k} \right]. \quad (\text{A17})$$

The left side minus the right side is a strictly convex quadratic in B with a negative constant term. It therefore has exactly one positive root. Direct substitution, using

$\binom{D-1}{k} = \frac{D-k}{D} \binom{D}{k}$, shows that this root is

$$B = B_k(m) = \frac{D \binom{n-1}{k} / \binom{D}{k} - D + k}{m}. \quad (\text{A18})$$

Thus (A5) is algebraically equivalent to the positive-root formula obtained from (A17), but is considerably simpler.

Substituting $x_l = B_k x_d$ into the disloyal first-order condition gives

$$(x_d^*)^2 = \frac{V \binom{D}{k} (m B_k + D - k)}{\binom{n-1}{k-1} D (m B_k + D)^2}. \quad (\text{A19})$$

From (A5),

$$m B_k + D = Q_k \quad \text{and} \quad m B_k + D - k = D R_k. \quad (\text{A20})$$

Moreover, (A7) implies

$$\frac{\binom{D}{k} D R_k}{\binom{n-1}{k-1} D} = \frac{\binom{n-1}{k}}{\binom{n-1}{k-1}} = \frac{n-k}{k}. \quad (\text{A21})$$

Equations (A19)–(A21) therefore give

$$x_d^* = \frac{E^*(k)}{Q_k}, \quad x_l^* = B_k \frac{E^*(k)}{Q_k}, \quad (\text{A22})$$

which proves part 1.

Now suppose $D < k < n$. Every top- k set necessarily contains at least one loyal agent. A disloyal agent consequently has zero probability of receiving the prize for every effort choice. Since effort is costly, $x_d^* = 0$. Conditional on type symmetry among loyal agents, differentiation of a loyal agent's winning probability gives

$$\frac{\partial P_{li}}{\partial x_{li}} = \frac{\sum_{j=0}^D \binom{D}{j} \binom{m-1}{k-j}}{\binom{n-1}{k-1} m^2 x_l}. \quad (\text{A23})$$

Vandermonde's identity reduces the sum in the numerator to $\binom{n-1}{k}$. Hence the loyal first-order condition is

$$x_l^2 = \frac{V}{m^2} \frac{\binom{n-1}{k}}{\binom{n-1}{k-1}} = \frac{V(n-k)}{m^2 k}, \quad (\text{A24})$$

which establishes (A10).

Finally, if $k = n$, every subordinate is eligible independently of performance. Winning probabilities are therefore locally and globally invariant to effort. The unique cost-minimizing choice of each subordinate is zero, proving part 3. $\square$

Remark A.1 (Boundary populations). *If $m = 0$, all agents are disloyal; if $m = n$,*

all are loyal. In either case the standard symmetric multiple-winner contest gives individual effort $E^*(k)/n$ for $k < n$, aggregate effort $E^*(k)$, and zero effort for $k = n$. Expressions involving the ratio $B_k = x_l/x_d$ are, of course, inapplicable when one type is absent.

Effort asymmetry and effort collapse

Lemma A.2 (Effort asymmetry under stochastic favoritism). *Suppose $1 \leq m \leq n - 1$. In the symmetric equilibrium,*

$$x_l^*(k; m) \geq x_d^*(k; m). \quad (\text{A25})$$

Equality holds at $k = 1$ and $k = n$. For every intermediate rule $1 < k < n$, the inequality is strict.

Proof. First suppose $k \leq D$. From Lemma A.1, $x_l^*/x_d^* = B_k$. At $k = 1$,

$$R_1 = \frac{n-1}{D}, \quad B_1 = \frac{D(n-1)/D - D + 1}{m} = 1. \quad (\text{A26})$$

For $k > 1$, use Vandermonde's identity to write

$$\binom{n-1}{k} = \sum_{r=0}^{\min\{m-1, k\}} \binom{m-1}{r} \binom{D}{k-r}. \quad (\text{A27})$$

The $r = 0$ and $r = 1$ terms alone imply

$$R_k \geq 1 + (m-1) \frac{\binom{D}{k-1}}{\binom{D}{k}} = 1 + (m-1) \frac{k}{D-k+1}. \quad (\text{A28})$$

Substituting this bound into (A5) gives $B_k > 1$ for $1 < k \leq D$. Hence $x_l^* > x_d^* > 0$ in this region.

If $D < k < n$, Lemma A.1 gives $x_l^* = E^*(k)/m > 0 = x_d^*$. If $k = n$, both efforts equal zero. Combining the three cases proves the result. $\square$

Corollary A.1 (Effort collapse under degenerate competition). *Suppose $D < k$. If $k < n$, disloyal agents exert zero effort while loyal agents exert $E^*(k)/m > 0$. If $k = n$, both types exert zero effort.*

Proof. Immediate from parts 2 and 3 of Lemma A.1. $\square$

Aggregate effort

Proposition A.1 (Aggregate-output invariance). *For every $n \geq 2$, $V > 0$, $m \in \{0, \dots, n\}$, and $k \in \mathcal{K}$, aggregate equilibrium effort satisfies*

$$mx_l^*(k; m) + Dx_d^*(k; m) = E^*(k) = \sqrt{\frac{V(n-k)}{k}}. \quad (\text{A29})$$

Consequently, aggregate effort is independent of m conditional on k and is strictly decreasing in k for $k < n$.

Proof. If $k \leq D$, equations (A9) and (A20) imply

$$mx_l^*(k; m) + Dx_d^*(k; m) = (mB_k + D) \frac{E^*(k)}{Q_k} = E^*(k). \quad (\text{A30})$$

If $D < k < n$, equation (A10) gives

$$mx_l^*(k; m) + Dx_d^*(k; m) = m \frac{E^*(k)}{m} + D \cdot 0 = E^*(k). \quad (\text{A31})$$

At $k = n$, both sides of (A29) equal zero. The same formula applies in the one-type boundary populations described above.

Independence from m follows directly from (A29). To establish monotonicity in k , observe that

$$E^*(k)^2 = V \left(\frac{n}{k} - 1 \right), \quad (\text{A32})$$

which is strictly decreasing in k . Since $E^*(k) \geq 0$, $E^*(k)$ itself is strictly decreasing on $\{1, \dots, n\}$. $\square$

The probability of a disloyal winner

Lemma A.3 (Equilibrium probability of a disloyal winner). *Suppose $1 \leq m \leq n - 1$. In equilibrium,*

$$P_d^*(k; m) = \begin{cases} \frac{n-k}{R_k(m) [k + DR_k(m)]}, & k \leq D, \\ 0, & k > D. \end{cases} \quad (\text{A33})$$

Moreover, $P_d^(k; m)$ is strictly decreasing in k on $\{1, \dots, D+1\}$ and remains zero thereafter.*

Proof. For $k \leq D$, the aggregate probability that a disloyal agent wins is

$$P_d^*(k; m) = \frac{D \binom{D-1}{k-1} x_d^*}{\binom{n-1}{k-1} (Dx_d^* + mx_l^*)}. \quad (\text{A34})$$

Using (A8), $x_d^*/E^*(k) = 1/Q_k$, and $\binom{n-1}{k-1} = \frac{k}{n-k} \binom{n-1}{k}$, equation (A34) becomes

$$\begin{aligned} P_d^*(k; m) &= \frac{k \binom{D}{k}}{\binom{n-1}{k-1} Q_k} \\ &= \frac{n-k}{R_k(k + DR_k)}, \end{aligned} \quad (\text{A35})$$

which is (A33). If $k > D$, every eligible set contains a loyal agent, so stochastic favoritism makes the probability of a disloyal winner zero.

It remains to prove monotonicity. For $k < D$,

$$\frac{R_{k+1}}{R_k} = \frac{n-1-k}{D-k} = 1 + \frac{m-1}{D-k} \geq 1. \quad (\text{A36})$$

Thus, as k rises, the numerator $n-k$ in (A33) strictly decreases, while both factors in its denominator weakly increase and the factor $k + DR_k$ strictly increases. Therefore $P_d^*(k+1; m) < P_d^*(k; m)$ for $k < D$. Finally, $P_d^*(D; m) > 0 = P_d^*(D+1; m)$, completing the proof. $\square$

Two useful boundary values follow immediately:

$$P_d^*(1; m) = \frac{D}{n} = \frac{n-m}{n}, \quad P_d^*(k; m) = 0 \quad \text{for all } k > D. \quad (\text{A37})$$

The superior's optimal competition rule

Substituting equilibrium effort into the superior's payoff gives the reduced problem

$$U_S(k; \alpha, m) = E^*(k) + \alpha A(k; m), \quad A(k; m) \equiv 1 - 2P_d^*(k; m), \quad (\text{A38})$$

which is the closed-form counterpart of the main text's eq. (1), with $A(k; m)$ the same expected-authority-value quantity defined there. Lemma A.3 implies that $A(k; m)$ is weakly increasing in k and strictly increasing until it reaches its upper bound of one.

Because the maximizer need not be unique at switching values, define the optimal correspondence

$$K^*(\alpha, m) \equiv \arg \max_{k \in \mathcal{K}} U_S(k; \alpha, m), \quad (\text{A39})$$

and its smallest and largest selections by

$$\underline{k}^*(\alpha, m) = \min K^*(\alpha, m), \quad \bar{k}^*(\alpha, m) = \max K^*(\alpha, m). \quad (\text{A40})$$

Lemma A.4 (Benchmark without authority concerns). *If $\alpha = 0$, the superior uniquely chooses $k^* = 1$.*

Proof. At $\alpha = 0$, equation (A38) reduces to $U_S(k; 0, m) = E^*(k)$. Proposition A.1 estab-

lishes that $E^*(k)$ is strictly decreasing in k . Hence its unique maximizer is $k = 1$. $\square$

Proposition A.2 (Optimal competition and legal weakness). *For fixed n , V , and m , both $\underline{k}^*(\alpha, m)$ and $\bar{k}^*(\alpha, m)$ are weakly increasing in α . Equivalently, weaker legal enforcement, represented by a larger α , cannot induce the superior to choose a strictly more intense competition rule.*

Proof. Let $k' > k$ and $\alpha' > \alpha$. From (A38),

$$\begin{aligned} & [U_S(k'; \alpha', m) - U_S(k; \alpha', m)] - [U_S(k'; \alpha, m) - U_S(k; \alpha, m)] \\ & = (\alpha' - \alpha)[A(k'; m) - A(k; m)] \geq 0, \end{aligned} \quad (\text{A41})$$

where the inequality follows from Lemma A.3. Thus U_S has increasing differences in (k, α) . Since $\mathcal{K}$ is a finite lattice, the monotone-selection theorem implies that both the smallest and largest elements of the argmax correspondence are weakly increasing in α .

For completeness, the same result follows directly. Choose $k \in K^*(\alpha, m)$ and $k' \in K^*(\alpha', m)$. Optimality gives

$$U_S(k; \alpha, m) \geq U_S(k'; \alpha, m), \quad (\text{A42})$$

$$U_S(k'; \alpha', m) \geq U_S(k; \alpha', m). \quad (\text{A43})$$

Adding these inequalities yields

$$(\alpha' - \alpha)[A(k'; m) - A(k; m)] \geq 0. \quad (\text{A44})$$

If $k' < k$, monotonicity of A reverses the weak inequality, and strict monotonicity over the payoff-relevant region rules out such a reversal except among payoff-equivalent choices after A has reached one; among those choices E^* selects the smallest k . Hence the extremal optimal selections cannot decrease with α . $\square$

Remark A.2 (Correct global switching conditions). *Adjacent comparisons alone do not generally characterize the global optimum. For any $\ell > k$ with $A(\ell; m) > A(k; m)$, define the pairwise cutoff*

$$\alpha_{k\ell}(m) \equiv \frac{E^*(k) - E^*(\ell)}{A(\ell; m) - A(k; m)} > 0. \quad (\text{A45})$$

Then ℓ weakly dominates k if and only if $\alpha \geq \alpha_{k\ell}(m)$. A rule k for which no smaller rule has the same value of A is globally optimal if and only if it beats every alternative:

$$\max_{j < k: A(k; m) > A(j; m)} \alpha_{jk}(m) \leq \alpha \leq \min_{\ell > k: A(\ell; m) > A(k; m)} \alpha_{k\ell}(m), \quad (\text{A46})$$

with the usual conventions for an empty maximum or minimum. If the interval in (A46) is empty, k is never optimal. This formulation allows the optimal rule to jump over one or more values of k . It does not require the adjacent cutoffs to be ordered. If some $j < k$ satisfies $A(j; m) = A(k; m)$, then $E^*(j) > E^*(k)$ and j strictly dominates k for every α ; such a k is never optimal.

Corollary A.2 (Induced productivity effect of legal weakness). *For fixed n, V , and m , equilibrium aggregate output under any monotone selection of the optimal competition correspondence is weakly decreasing in α .*

Proof. Proposition A.2 establishes that the selected k^* is weakly increasing in α , while Proposition A.1 establishes that $E^*(k)$ is strictly decreasing in k . Their composition $E^*(k^*(\alpha, m))$ is therefore weakly decreasing in α . $\square$

Effort gaps

Proposition A.3 (Political origin and incidence of effort gaps). *Suppose both types are present, $1 \leq m \leq n-1$. Conditional on a competition rule k , the equilibrium effort gap is*

$$\Gamma(k; m) \equiv x_l^*(k; m) - x_d^*(k; m) = \begin{cases} \frac{[B_k(m) - 1]E^*(k)}{Q_k(m)}, & k \leq D, \\ \frac{E^*(k)}{m}, & D < k < n, \\ 0, & k = n. \end{cases} \quad (\text{A47})$$

Consequently,

$$\Gamma(k; m) \begin{cases} = 0, & k \in \{1, n\}, \\ > 0, & 1 < k < n. \end{cases} \quad (\text{A48})$$

Thus effort inequality arises exactly when the competition rule leaves room for loyalty-based priority while retaining some performance dependence.

Proof. Equation (A47) follows by subtracting the two effort expressions in Lemma A.1. Lemma A.2 establishes that $B_1 = 1$, $B_k > 1$ for $1 < k \leq D$, and that loyal effort is strictly positive while disloyal effort is zero for $D < k < n$. At $k = n$, both types exert zero effort. These observations give (A48). $\square$

Remark A.3 (What is and is not established about “hump shape”). Equation (A48) proves that the effort gap vanishes at both extremes and is positive at every intermediate rule. That fact alone does not logically prove that the discrete sequence $\Gamma(1; m), \dots, \Gamma(n; m)$ is single-peaked. A claim of strict single-peakedness requires an additional compar-

ison of $\Gamma(k+1; m)$ and $\Gamma(k; m)$ using (A47). Unless that additional inequality is supplied, the general statement should be phrased as “zero at the extremes and positive at intermediate competition rules,” rather than as a formally proven single-peakedness result.

Political inequality in the final equilibrium

Corollary A.3 (Political Inequality in the Final Equilibrium). $\mathcal{I}^*(\alpha, m) > 0$ if and only if $1 < k^*(\alpha, m) < n$.

Proof. By Lemma A.2 (equivalently, eq. (A48)), $I^*(k; m) \equiv x_l^*(k; m) - x_d^*(k; m) = 0$ exactly when $k \in \{1, n\}$, and $I^*(k; m) > 0$ for every $1 < k < n$. Since $k^*(\alpha, m) \in \{1, \dots, n\}$ by construction, substituting $k = k^*(\alpha, m)$ gives $\mathcal{I}^*(\alpha, m) \equiv I^*(k^*(\alpha, m); m) > 0$ if and only if $k^*(\alpha, m) \notin \{1, n\}$, i.e., $1 < k^*(\alpha, m) < n$. $\square$

Authority capital at the extremes of reliance on loyalty

Proposition 3 describes three regimes through which authority capital m affects the optimal eligibility rule: neutrality, discretion activation, and authority–discretion substitution. The two extremes of reliance on loyalty admit general proofs, valid for every n and V ; only the intermediate activation region resists a single closed-form characterization, as the counterexample below confirms.

Proposition A.4 (Neutrality at low reliance on loyalty). Suppose $1 \leq m \leq n - 1$. Define

$$\alpha_N(n, V) \equiv \min_{1 \leq m \leq n-1} \min_{2 \leq \ell \leq n} \alpha_{1\ell}(m), \quad \alpha_{1\ell}(m) \equiv \frac{E^*(1) - E^*(\ell)}{A(\ell; m) - A(1; m)}. \quad (\text{A49})$$

Then $\alpha_N(n, V) > 0$, and for every $\alpha \in [0, \alpha_N(n, V)]$ and every $1 \leq m \leq n - 1$, $k = 1 \in K^*(\alpha, m)$; if $\alpha < \alpha_N(n, V)$, $k^*(\alpha, m) = 1$ is the unique optimum. Authority-capital neutrality therefore holds not merely in the limit $\alpha \rightarrow 0$, but uniformly over the entire interval $[0, \alpha_N(n, V))$, with $\alpha_N(n, V) > 0$ independent of m .

Proof. By Lemma A.3, $A(k; m)$ is nondecreasing in k and strictly increasing on $\{1, \dots, D+1\}$, where $D = n - m \geq 1$; since $A(1; m) = 1 - 2(n - m)/n < 1$ for $m \geq 1$ and $A(k; m)$ reaches its maximum value of one at $k = D + 1 \leq n$, it follows that $A(\ell; m) > A(1; m)$ for every $\ell \geq 2$. Because E^* is strictly decreasing (Proposition A.1), $\alpha_{1\ell}(m)$ in (A49) is therefore well defined, finite, and strictly positive for every $\ell \in \{2, \dots, n\}$ and every $m \in \{1, \dots, n - 1\}$. This index set is finite, so $\alpha_N(n, V)$, being the minimum of finitely many strictly positive numbers, is itself strictly positive.

Fix m and $\alpha \leq \alpha_N(n, V)$. There is no $j < 1$ in $\mathcal{K}$, so the lower bound of the global-optimality interval in Remark A.2 is vacuous, and $k = 1$ is optimal if and only if $\alpha \leq$

$\min_{\ell > 1} \alpha_{1\ell}(m)$. Since $\alpha_N(n, V) \leq \alpha_{1\ell}(m)$ for every such ℓ , this holds, so $1 \in K^*(\alpha, m)$. If $\alpha < \alpha_N(n, V)$, the inequality is strict against every $\ell > 1$, so Remark A.2 implies $k = 1$ strictly dominates every $\ell > 1$, giving $K^*(\alpha, m) = \{1\}$. As m was arbitrary, the claim holds for every $1 \leq m \leq n - 1$. $\square$

Proposition A.5 (Substitution at high reliance on loyalty). *Suppose $1 \leq m \leq n - 1$ and let $D = n - m$. By Lemma A.3, $A(k; m)$ first attains its maximum value of one at $k^\dagger \equiv D + 1 = n - m + 1$. Define, for each $k' \in \{1, \dots, D\}$,*

$$\alpha_{k'}^\dagger(m) \equiv \frac{E^*(k') - E^*(n - m + 1)}{1 - A(k'; m)} > 0, \quad \alpha_S(n, V) \equiv \max_{1 \leq m \leq n-1} \max_{1 \leq k' \leq n-m} \alpha_{k'}^\dagger(m) < \infty. \quad (\text{A50})$$

Then for every $\alpha > \alpha_S(n, V)$ and every $1 \leq m \leq n - 1$,

$$k^*(\alpha, m) = n - m + 1, \quad (\text{A51})$$

which is strictly decreasing in m . Consequently, for sufficiently weak legal enforcement, authority capital and promotion discretion are global substitutes: a marginal increase in m strictly tightens the optimal eligibility rule and strictly raises equilibrium output $\hat{E}(\alpha, m) = E^(n - m + 1)$.*

Proof. Write $k^\dagger \equiv n - m + 1$. For every $k'' > k^\dagger$, Lemma A.3 gives $A(k''; m) = A(k^\dagger; m) = 1$, so

$$U_S(k^\dagger; \alpha, m) - U_S(k''; \alpha, m) = E^*(k^\dagger) - E^*(k'') > 0$$

for every $\alpha \geq 0$, by strict monotonicity of E^* (Proposition A.1); $k^\dagger$ therefore strictly dominates every larger rule regardless of α .

For $k' \leq D$, $A(k'; m) < 1 = A(k^\dagger; m)$, so $\alpha_{k'}^\dagger(m)$ in (A50) coincides with the pairwise cutoff $\alpha_{k'k^\dagger}(m)$ of Remark A.2, and $k^\dagger$ weakly dominates k' once $\alpha \geq \alpha_{k'}^\dagger(m)$, strictly once $\alpha > \alpha_{k'}^\dagger(m)$. Since $\alpha_S(n, V) \geq \alpha_{k'}^\dagger(m)$ for every $k' \leq D$ and every m , any $\alpha > \alpha_S(n, V)$ has $k^\dagger$ strictly dominating every $k' < k^\dagger$ as well as every $k'' > k^\dagger$, so $K^*(\alpha, m) = \{n - m + 1\}$. Because $D = n - m$ is strictly decreasing in m , so is $k^\dagger = n - m + 1$; the output claim follows from $\hat{E}(\alpha, m) \equiv E^*(k^*(\alpha, m))$ together with strict monotonicity of E^* . $\square$

Between the two thresholds identified above, $\alpha \in (\alpha_N(n, V), \alpha_S(n, V)]$, no general monotonicity claim in m survives. The following counterexample confirms this and shows that Proposition A.2 — not any claim about m — is the strongest globally valid comparative-static result available for the baseline model.

Proposition A.6 (Counterexample to monotonicity in m). *The optimal competition rule need not be monotone in authority capital m . In particular, let $n = 5$, $V = 1$, and*

$\alpha = 1$. Then

$$k^*(1, 1) = 1, \quad k^*(1, 2) = 2, \quad k^*(1, 3) = 1. \quad (\text{A52})$$

Proof. Using (A38) and Lemma A.3, the relevant payoffs are as follows. When $m = 1$,

$$\begin{aligned} U_S(1; 1, 1) &= 1.4, & U_S(2; 1, 1) &= \sqrt{\frac{3}{2}} \approx 1.2247, \\ U_S(3; 1, 1) &\approx 1.2451, & U_S(4; 1, 1) &= 1.25, & U_S(5; 1, 1) &= 1, \end{aligned} \quad (\text{A53})$$

so $k^*(1, 1) = 1$. When $m = 2$,

$$U_S(1; 1, 2) = 1.8, \quad U_S(2; 1, 2) = \sqrt{\frac{3}{2}} + \frac{5}{8} \approx 1.8497, \quad (\text{A54})$$

and the remaining rules yield smaller payoffs, so $k^*(1, 2) = 2$. When $m = 3$,

$$U_S(1; 1, 3) = 2.2, \quad U_S(2; 1, 3) = \sqrt{\frac{3}{2}} + \frac{13}{14} \approx 2.1533, \quad (\text{A55})$$

and the remaining rules again yield smaller payoffs, so $k^*(1, 3) = 1$. Therefore k^* moves from 1 to 2 and then back to 1 as m rises, which contradicts both weakly increasing and weakly decreasing global monotonicity. $\square$

Authority capital thus affects the superior's optimal competition rule through two opposing margins. At intermediate levels, a larger loyal group can increase the marginal political return to relaxing performance competition; at high levels, it can make strict competition compatible with a high probability of a loyal winner. Consequently, the effect of m on k^* is generally non-monotonic between the thresholds $\alpha_N(n, V)$ and $\alpha_S(n, V)$ identified above, exactly as Proposition 3 states, and any claim that equilibrium productivity is monotone in m through k^* on that intermediate range requires additional assumptions or a modification of the promotion technology.

Productivity reversals across environments

Proposition A.7 (Higher authority reliance can sustain higher output). *For every $n \geq 3$ and $V > 0$, there exist two baseline environments (α_L, m_L) and (α_H, m_H) satisfying*

$$\alpha_H > \alpha_L \quad \text{and} \quad m_H > m_L,$$

such that

$$k^*(\alpha_H, m_H) < k^*(\alpha_L, m_L) \quad \text{and consequently} \quad \widehat{E}(\alpha_H, m_H) > \widehat{E}(\alpha_L, m_L).$$

Thus, although a larger α weakly relaxes competition conditional on a fixed level of authority capital, legal weakness does not impose an unconditional ordering on pro-

ductivity across environments with different stocks of authority capital.

Proof. Recall $E^*(k) = \sqrt{V(n-k)/k}$ and $A(k; m) = 1 - 2P_d^*(k; m)$, so that $U_S(k; \alpha, m) = E^*(k) + \alpha A(k; m)$ by (A38).

We construct two environments with $m_L = 1$ and $m_H = n - 1$.

Consider first the low-authority-capital environment $m_L = 1$. Under the strict tournament $k = 1$, $P_d^*(1; 1) = \frac{n-1}{n}$, and hence

$$A(1; 1) = 1 - \frac{2(n-1)}{n} = \frac{2-n}{n}.$$

Now consider the weaker competition rule $k = n - 1$. Since $m = 1$, the number of disloyal subordinates is $D = n - 1$, and the closed-form winning probability gives $P_d^*(n-1; 1) = \frac{1}{2(n-1)}$. Therefore,

$$A(n-1; 1) = 1 - \frac{1}{n-1} = \frac{n-2}{n-1}.$$

The superior prefers $k = n - 1$ to $k = 1$ whenever

$$E^*(n-1) + \alpha A(n-1; 1) > E^*(1) + \alpha A(1; 1).$$

Equivalently, $\alpha > \hat{\alpha}_L(n, V)$, where

$$\hat{\alpha}_L(n, V) \equiv \frac{E^*(1) - E^*(n-1)}{A(n-1; 1) - A(1; 1)}.$$

Using $E^*(1) = \sqrt{V(n-1)}$ and $E^*(n-1) = \sqrt{V/(n-1)}$, we obtain

$$E^*(1) - E^*(n-1) = \sqrt{V} \frac{n-2}{\sqrt{n-1}}.$$

Moreover,

$$A(n-1; 1) - A(1; 1) = \frac{n-2}{n-1} + \frac{n-2}{n} = \frac{(n-2)(2n-1)}{n(n-1)}.$$

Hence

$$\hat{\alpha}_L(n, V) = \frac{n\sqrt{V(n-1)}}{2n-1}.$$

It follows that, for every $\alpha_L > \hat{\alpha}_L(n, V)$, the strict tournament $k = 1$ is not optimal when $m_L = 1$, because $k = n - 1$ yields a strictly higher payoff. Therefore, $k^*(\alpha_L, 1) > 1$.

Now consider the high-authority-capital environment $m_H = n - 1$. There is only one disloyal subordinate. Under $k = 1$, $P_d^*(1; n-1) = \frac{1}{n}$, so $A(1; n-1) = 1 - \frac{2}{n}$. For every $k \geq 2$, the eligible set necessarily contains at least one loyal subordinate. Thus

$P_d^*(k; n-1) = 0$ and $A(k; n-1) = 1$.

Among all $k \geq 2$, the rule $k = 2$ produces the highest aggregate output because $E^*(k)$ is strictly decreasing in k . Therefore, to establish that $k = 1$ is globally optimal, it is sufficient to compare $k = 1$ with $k = 2$.

The superior prefers $k = 1$ to $k = 2$ whenever

$$E^*(1) + \alpha A(1; n-1) > E^*(2) + \alpha A(2; n-1).$$

Equivalently, $\alpha < \hat{\alpha}_H(n, V)$, where

$$\hat{\alpha}_H(n, V) \equiv \frac{E^*(1) - E^*(2)}{A(2; n-1) - A(1; n-1)}.$$

Since

$$A(2; n-1) - A(1; n-1) = 1 - \left(1 - \frac{2}{n}\right) = \frac{2}{n},$$

we obtain

$$\hat{\alpha}_H(n, V) = \frac{n}{2} \left[\sqrt{V(n-1)} - \sqrt{\frac{V(n-2)}{2}} \right].$$

We next show that the interval $(\hat{\alpha}_L(n, V), \hat{\alpha}_H(n, V))$ is nonempty. It is sufficient to prove $\hat{\alpha}_H(n, V) > \hat{\alpha}_L(n, V)$. After dividing both sides by the positive term $n\sqrt{V}$, this inequality becomes

$$\frac{1}{2} \left[\sqrt{n-1} - \sqrt{\frac{n-2}{2}} \right] > \frac{\sqrt{n-1}}{2n-1}.$$

Equivalently,

$$(2n-3)\sqrt{n-1} > (2n-1)\sqrt{\frac{n-2}{2}}.$$

Both sides are nonnegative for $n \geq 3$, so squaring preserves the inequality. The resulting condition is

$$2(2n-3)^2(n-1) > (2n-1)^2(n-2).$$

The difference between the left-hand side and the right-hand side is

$$F(n) = 4n^3 - 20n^2 + 33n - 16.$$

At $n = 3$, $F(3) = 11 > 0$. Moreover,

$$F(n+1) - F(n) = 12n^2 - 28n + 17 > 0$$

for every integer $n \geq 3$. Hence $F(n) > 0$ for every $n \geq 3$, establishing $\hat{\alpha}_H(n, V) > \hat{\alpha}_L(n, V)$.

Because the interval is nonempty, choose any $\hat{\alpha}_L(n, V) < \alpha_L < \alpha_H < \hat{\alpha}_H(n, V)$.

In the low-authority-capital environment $(\alpha_L, 1)$, $k^*(\alpha_L, 1) > 1$. In the high-authority-capital environment $(\alpha_H, n - 1)$, $k^*(\alpha_H, n - 1) = 1$. Therefore,

$$k^*(\alpha_H, n - 1) < k^*(\alpha_L, 1),$$

even though $\alpha_H > \alpha_L$.

Finally, because aggregate equilibrium output $E^*(k) = \sqrt{V(n - k)/k}$ is strictly decreasing in k , it follows that $\widehat{E}(\alpha_H, n - 1) > \widehat{E}(\alpha_L, 1)$. This proves the result. $\square$

Two-Period Extension: Institutional Adjustment and Authority Accumulation

The inherited state is (m_0, α_0) , with $m_0 \in \{0, \dots, n\}$ and $\alpha_0 \in [0, \infty)$. Recall from the main text that legal enforcement of effectiveness $\ell(\alpha)$ enters the superior's flow payoff, with benchmark specification $\ell(\alpha) = 1/(1 + \alpha)$, so that

$$u(k, m, \alpha) = E^*(k) + \alpha A(k; m) + \beta \ell(\alpha), \quad \beta > 0, \quad (\text{A56})$$

and $\widetilde{W}(m, \alpha) \equiv \max_{k \in \mathcal{K}} u(k, m, \alpha)$. After period 0, the superior chooses $z \in \{-m_0, \dots, n - m_0\}$, so $m_1 = m_0 + z$; there is no depreciation. In period 1 she chooses $(\alpha_1, k_1) \in [0, \infty) \times \mathcal{K}$.

The direction-specific authority adjustment cost is

$$C_M(z; \alpha_0) = \begin{cases} \frac{\gamma_+}{2(1 + \eta_M \alpha_0)} z^2, & z > 0, \\ 0, & z = 0, \\ \frac{\gamma_-}{2} z^2, & z < 0, \end{cases} \quad \gamma_+ > 0, \gamma_- > 0, \eta_M \geq 0, \quad (\text{A57})$$

where $z \equiv m_1 - m_0$. Because $\partial C_M / \partial \alpha_0 = -\gamma_+ \eta_M z^2 / [2(1 + \eta_M \alpha_0)^2] \leq 0$ for $z > 0$, expanding the loyal network is cheaper the weaker is the inherited legal environment α_0 ; the $z < 0$ branch carries no α_0 -dependence, consistent with Condition 1. Since $m_0, m_1 \in \{0, 1, \dots, n\}$, the relevant comparison is always over the finite set $z \in \{-m_0, \dots, n - m_0\}$, not a continuous first-order condition.

The legal-strength adjustment cost is

$$C_\alpha(a; m) = \begin{cases} F_S(m) + \frac{c_S(m)}{2} a^2, & a < 0, \\ 0, & a = 0, \\ F_W(m) + \frac{c_W(m)}{2} a^2, & a > 0, \end{cases} \quad (\text{A58})$$

where all fixed costs are nonnegative and $c_S(m), c_W(m) > 0$; F_S, c_S are weakly increas-

ing in m , while F_W, c_W are weakly decreasing in m , consistent with Condition 1's second sentence.

The complete objective is

$$\begin{aligned} \Pi_S = & \widetilde{W}(m_0, \alpha_0) - C_M(z; \alpha_0) \\ & + \delta [u(k_1, m_0 + z, \alpha_1) - C_\alpha(\alpha_1 - \alpha_0; m_0 + z)], \quad \delta \in (0, 1]. \end{aligned} \quad (\text{A59})$$

Lemma A.5 (Marginal institutional incentive). *Holding (k, m) fixed,*

$$\frac{\partial u}{\partial \alpha} = A(k; m) - \frac{\beta}{(1 + \alpha)^2}. \quad (\text{A60})$$

If $A(k; m) > 0$, the derivative changes sign at

$$\widehat{\alpha}(k, m) = \max \left\{ 0, \sqrt{\frac{\beta}{A(k; m)}} - 1 \right\}. \quad (\text{A61})$$

If $A(k; m) \leq 0$, it is negative for every $\alpha \geq 0$.

Proof. Differentiate (A56) using $\ell(\alpha) = 1/(1 + \alpha)$. When $A(k; m) > 0$, solving $A(k; m) = \beta/(1 + \alpha)^2$ yields (A61); the derivative is strictly increasing in α . When $A(k; m) \leq 0$, both terms in (A60) are nonpositive and the second is strictly negative. $\square$

For a given (m, α_0) define

$$S(m, \alpha_0) = \max_{\substack{k \in \mathcal{K} \\ 0 \leq \alpha < \alpha_0}} \{u(k, m, \alpha) - C_\alpha(\alpha - \alpha_0; m)\}, \quad (\text{A62})$$

$$Q(m, \alpha_0) = \widetilde{W}(m, \alpha_0), \quad (\text{A63})$$

$$D(m, \alpha_0) = \max_{\substack{k \in \mathcal{K} \\ \alpha > \alpha_0}} \{u(k, m, \alpha) - C_\alpha(\alpha - \alpha_0; m)\}, \quad (\text{A64})$$

$$J(m; \alpha_0) = \max\{S(m, \alpha_0), Q(m, \alpha_0), D(m, \alpha_0)\}. \quad (\text{A65})$$

Proposition A.8 (Institutional adjustment regions). *With ties resolved in favor of the status quo, the period-1 choice satisfies*

$$\alpha_1^* \begin{cases} < \alpha_0, & S > \max\{Q, D\}, \\ = \alpha_0, & Q \geq \max\{S, D\}, \\ > \alpha_0, & D > \max\{S, Q\}. \end{cases} \quad (\text{A66})$$

Equivalently, define gross gains net of variable costs by

$$G_S = \max_{k, 0 \leq \alpha < \alpha_0} \left\{ u(k, m, \alpha) - \frac{c_S(m)}{2} (\alpha - \alpha_0)^2 \right\} - Q, \quad (\text{A67})$$

$$G_W = \max_{k, \alpha > \alpha_0} \left\{ u(k, m, \alpha) - \frac{c_W(m)}{2} (\alpha - \alpha_0)^2 \right\} - Q. \quad (\text{A68})$$

Then status quo is optimal if and only if

$$G_S(m, \alpha_0) \leq F_S(m), \quad G_W(m, \alpha_0) \leq F_W(m). \quad (\text{A69})$$

Proof. The three feasible subsets $\alpha < \alpha_0$, $\alpha = \alpha_0$, and $\alpha > \alpha_0$ are mutually exclusive and exhaustive. Maximizing within each subset produces S, Q, D , so their maximum gives (A66). Because the appropriate fixed cost is paid exactly when the superior leaves the status quo, $S - Q = G_S - F_S$ and $D - Q = G_W - F_W$. Hence Q weakly dominates both alternatives exactly under (A69). $\square$

Proposition A.9 (Subgame-perfect equilibrium). *A pure-strategy subgame-perfect equilibrium exists. It is characterized by*

$$(\alpha_1^*(m), k_1^*(m)) \in \arg \max_{\alpha \geq 0, k \in \mathcal{K}} \{u(k, m, \alpha) - C_\alpha(\alpha - \alpha_0; m)\}, \quad (\text{A70})$$

$$z^*(m_0, \alpha_0) \in \arg \max_{z \in \{-m_0, \dots, n - m_0\}} \{-C_M(z; \alpha_0) + \delta J(m_0 + z; \alpha_0)\}. \quad (\text{A71})$$

The resulting equilibrium path is

$$(m_0, \alpha_0) \mapsto (m_1^*, \alpha_1^*, k_1^*), \quad m_1^* = m_0 + z^*. \quad (\text{A72})$$

The equilibrium is unique whenever both argmax sets are singletons; otherwise there are multiple payoff-equivalent SPE selections (or a unique selected SPE under an explicit tie-breaking rule).

Proof. For every fixed m and k , the positive quadratic adjustment term makes the period-1 objective tend to $-\infty$ as $\alpha \rightarrow \infty$. It is continuous within each adjustment region and has well-defined one-sided values at the status quo. Since $\mathcal{K}$ is finite, the maximum in (A70) is attained. Therefore $J(m; \alpha_0)$ is defined for every feasible m . The period-0 authority choice is over the finite set $\{-m_0, \dots, n - m_0\}$, so the maximum in (A71) is also attained. Backward induction establishes sequential optimality in every subgame. The uniqueness statement follows from uniqueness of the two nested maximizers. $\square$

For discrete authority capital, define the value of moving from m to m' as

$$\Gamma(m' \mid m_0, \alpha_0) = -C_M(m' - m_0; \alpha_0) + \delta J(m'; \alpha_0). \quad (\text{A73})$$

Thus $m_1^* > m_0$, $m_1^* = m_0$, or $m_1^* < m_0$ according to which candidate maximizes (A73). In particular, $m_1^* = m_0$ if and only if

$$\delta[J(m'; \alpha_0) - J(m_0; \alpha_0)] \leq C_M(m' - m_0; \alpha_0) \quad \text{for every feasible } m'. \quad (\text{A74})$$

Remark A.4 (What is general and what is numerical). *Equations (A66), (A69), and (A74) are exact general boundaries. A statement that a boundary is globally upward sloping, downward sloping, or unique requires additional single-crossing conditions, which we supply next for the one comparative static the main text relies on. The colored regions in the calibrated Figure 5 are obtained by evaluating the global maximizations in (A70) and (A71) at every grid point; their exact step shape, and in particular the full ordering of strengthen/status-quo/weaken regions along α_0 , is a numerical implication of the reported parameters, not a claim proved below for every contest structure or parameterization.*

Assumption A.1 (Authority–institutional complementarity). *For every inherited α_0 , the net value of institutional weakening,*

$$H_W(m; \alpha_0) \equiv D(m, \alpha_0) - \max\{Q(m, \alpha_0), S(m, \alpha_0)\}, \quad (\text{A75})$$

is weakly increasing in authority capital: $H_W(m + 1; \alpha_0) \geq H_W(m; \alpha_0)$ for every $m \in \{0, \dots, n - 1\}$.

Proposition A.10 (Authority capital and institutional weakening). *Under Assumption A.1, the set of authority-capital levels at which weakening the law is optimal is an upper set: if weakening is optimal at (m, α_0) , it remains optimal at (m', α_0) for every $m' > m$. Consequently, whenever this set is nonempty, there exists a tipping point*

$$m^W(\alpha_0) \equiv \min\{m : H_W(m; \alpha_0) > 0\} \quad (\text{A76})$$

such that the superior weakens legal enforcement if and only if $m \geq m^W(\alpha_0)$ (with $m^W(\alpha_0) \equiv +\infty$ if the set is empty).

Proof. By (A66), weakening is optimal at m if and only if $D(m, \alpha_0) > \max\{Q(m, \alpha_0), S(m, \alpha_0)\}$, i.e., $H_W(m; \alpha_0) > 0$. Suppose this holds. By Assumption A.1, $H_W(m'; \alpha_0) \geq H_W(m; \alpha_0) > 0$ for every $m' > m$, so $D(m', \alpha_0) > \max\{Q(m', \alpha_0), S(m', \alpha_0)\}$ and weakening remains optimal at m' . The tipping point follows immediately from the definition of an upper set on $\{0, \dots, n\}$. $\square$

Remark A.5. *Assumption A.1 is consistent with the model's two underlying mechanisms: a larger loyal coalition raises the political return to reliance on authority ($A(k; m)$ weakly increasing in m , in the sense already established for fixed k) and lowers the cost of weakening formal legal constraints (F_W, c_W weakly decreasing in m in*

(A58)). Because the competition rule k is discrete and is re-optimized within each of S , Q , and D , these primitive mechanisms do not by themselves guarantee global monotonicity of H_W without further single-crossing structure on $A(k; m)$ and k^* of the kind ruled out, in general, by Proposition A.6. We therefore state the required property directly, as Assumption A.1, and verify it under the benchmark calibration reported in Figure 5; we do not claim it holds for every contest structure or parameterization, and Figure 5's further ordering of regions along α_0 is reported as a calibration feature rather than as an implication of this assumption.

Checklist for replication

For each inherited state (m_0, α_0) , a numerical implementation should:

1. compute (x_ℓ^*, x_d^*) , $E^*(k)$, $P_d^*(k; m)$, and $A(k; m)$ for every (k, m) using (A9)–(A38);
2. globally maximize the period-1 objective over $\alpha_1 \geq 0$ for every (k, m_1) , retaining the best strengthening, status-quo, and weakening values;
3. construct $J(m_1; \alpha_0)$ from (A65);
4. compare every feasible m_1 using (A73); and
5. record $(m_1^*, \alpha_1^*, k_1^*)$ and classify the state using (A66) and the sign of $m_1^* - m_0$.

*

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